Calibrating an agentic structural analysis and design program against published benchmarks

Pondus: 470 comparisons with textbooks, verification manuals and code worked examples

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Abstract

Language-model agents can now build, analyse and design building structures through the tools a program exposes, so the correctness of those tools sets the correctness of everything the agent produces. This paper reports the calibration of Pondus, an open-source program in which an agent and an engineer share one structural model, against reference values from textbooks, software verification manuals and published worked examples of the design codes. Every comparison passes through the program's own model, analysis wrapper and design code rather than through the analysis engine it links, because the engine had already been verified by its authors and the risk sits in the layer above it. The calibration covers linear and nonlinear frame analysis, slab load transfer, plates, modal and seismic analysis, time history, shear walls and wall-frame interaction under earthquake and dynamic loading, the HL-93 moving-load envelopes of a published continuous bridge design, stability, plastic collapse, moving loads, staged construction, section properties, AISC 360-16 and EN 1993-1-1 steel design, the EN 1992-1-1 cross-check, ACI 318-19 beams, slabs, columns and footings, the ACI 318-63 two-way slab coefficients, Wood–Armer reinforcement, timber and masonry. Of 470 comparisons over 229 cases, 430 agree within 0.5 % and 449 within 1 %, and each of the remaining 21 is explained by mesh convergence, by the conservative load-contour method for biaxial bending, by the rounded SI constants of ACI 318 itself or by the shear deformation of walls that Euler–Bernoulli references leave out. The calibration, with the closed-form tests that preceded it, found fourteen defects, six of them unconservative, including a base shear coefficient 41 % low for stiff structures, a stirrup shear resistance 66 % high and a tension rupture capacity 42 % high, and a shear-wall model whose rigid arms silently lost precision and raised a wall-frame frequency by 11.5 %. Thirteen of the fourteen sat in the code that turns the model into the inputs of a kernel or a check, and only one in a numerical kernel, which suggests where verification effort for agent-driven engineering software should go. A separate validation against the full-scale seven-storey wall building tested on the UCSD–NEES shake table reproduces the measured first period within 6.2 % and the second-mode frequency within 0.9 % with gross sections, and the ACI 318 effective stiffnesses bracket the periods measured after the earthquake tests.

Keywords structural analysis software · verification · benchmark · design codes · ACI 318 · AISC 360 · Eurocode · language-model agents

1 Introduction

Structural design software has always needed evidence that it computes what its documentation claims, and the profession has built that evidence from benchmark problems with known answers. Verification manuals of commercial programs, the NAFEMS benchmarks and the worked examples that code bodies publish alongside their standards give an engineer a way to check a program before trusting it with a real building. The arrival of language-model agents that drive engineering programs through tool interfaces such as the Model Context Protocol [1] raises the stakes of that check. An agent can assemble a model, run an analysis and size every member in seconds, but it accepts the numbers the tools return, so an error in a tool propagates through the whole design without the pause in which an engineer would have noticed that a result looked wrong.

Pondus is a native program in which an agent and an engineer work on one structural model. The agent calls the same commands as the engineer's user interface, the analysis runs on the Stabileo engine [2], and the design follows ACI 318-19 [3], AISC 360-16 [4] and EN 1993-1-1 [5] among other codes. Stabileo ships its own verification suite against textbooks and design examples, so the engine's kernels were not the main concern. The concern was the layer that Pondus adds above the engine: the code that splits members, converts loads, carries slab loads to beams, assembles combinations, maps sign conventions and turns analysis results into the inputs of each design check. That layer is ordinary application code, written quickly, and it is exactly the code an agent depends on without seeing.

This paper reports a calibration of that layer against published reference values. The word calibration is used here in its everyday engineering sense of checking an instrument against known standards, and it corresponds to code and solution verification in the vocabulary of ASME V&V 10 [6] and Oberkampf and Roy [7]. No parameter was fitted to the references, so agreement means that the program reproduces the reference and disagreement means a defect, a deliberate simplification or a difference between editions of a code. The paper describes the method in Section 3, reports the results by aspect in Sections 4 to 7, analyses the defects the calibration found in Section 8 and states what was not calibrated in Section 9.

2 The program and what needed calibration

Pondus is organised in four layers. The engine is Stabileo's `dedaliano-engine` at commit 7985e8c, vendored into the workspace with three marked source changes: a sign correction in the vertical Winkler foundation matrix, a sparse Cholesky test placed before the dense rank test that the plastic solver runs at every hinge event, which changes speed only, and a moving-load envelope kept at each end of an element (defect 14). The core library holds the model, the analysis wrapper, the design codes, the advanced analyses, the calculation report and the interchange formats. The MCP server exposes the core's commands to agents, and the desktop application exposes the same commands to the engineer through a ribbon, tables and a chat.

The analysis wrapper does more than pass the model to the engine. It splits every member at the nodes that lie on it, at every load break point and into at least eight elements, so each element carries a load that is linear along its length. It transfers slab loads to the edge beams along the 45° lines of each panel, or one way when the aspect ratio exceeds 2, or it meshes the slabs into MITC4 shell elements [8] joined to the frame. It builds the seismic mass from the model's own load cases, scales the response spectrum result to the equivalent lateral force base shear, generates each shear wall as a wide column with rigid arms at every level, routes nonlinear members and supports to the engine's contact, cable and Winkler solvers, and computes pushover, moving-load envelopes, influence lines and construction stages on models of its own. The design code is Pondus's own for ACI 318 beams, slabs, columns and footings, for AISC 360 members and for the slab coefficient methods, and it wraps the engine's checkers for EN 1993-1-1, the EN 1992-1-1 cross-check, the shear tab bolt and weld groups, timber and masonry.

Each of these steps can be wrong while the engine underneath is right, because each step decides what the engine or the checker is asked. A wrong effective length, a wrong axis order or a missing coefficient

produces a confident and plausible number. The calibration was therefore designed so that every comparison enters through the same functions the program uses when it designs a building.

3 Method

3.1 Reference values

Each comparison is recorded with its source, the quantity, its unit, the reference value, the Pondus value, the relative error, the tolerance and the kind of reference. A reference of kind *published* is a number printed in the cited source, such as the $\phi_c P_n = 892$ kips of AISC Design Example E.1C [9] or the coefficient $w_{max} = 0.00406\, qa^4/D$ of Timoshenko and Woinowsky-Krieger's Table 8 [10]. A reference of kind *closed-form* evaluates, in the test, a formula of the cited source, such as $6EI\Delta/L^2$ for a support settlement [11] or the CQC combination of Der Kiureghian [12], and the formula is named in the record. A reference of kind *independent* is an exact solution by another method, used for biaxial column capacity and for the flexural–shear continuum of a wall coupled to frames, whose frequency equation the test solves itself. Of the 470 comparisons, 331 are published, 118 closed-form and 21 independent.

The published values come from four groups of sources. The first is the set of inputs and printed answers that Stabileo records, with page numbers, from the AISC Design Examples v16 [9], the SCI/CTICM Steel Buildings in Europe examples [13], SCI P364 [14], the JRC Eurocode 2 worked examples [15] and the INTI-CIRSOC examples [16, 17]. The second is the classical textbook literature of structural analysis, dynamics, stability and plates [10, 11, 18–30]. The third is the ANSYS Verification Manual as reproduced by PyMAPDL [31]. The fourth is a set of ACI 318 worked examples published by StructurePoint after Wight [32] and other textbooks [33], a scan of the ACI 318-63 Method 2 coefficient table in lecture notes of the University of Basrah [34, 35], and a timber column example by Dlubal [36, 37].

3.2 The rule that everything passes through Pondus

No comparison calls the engine's checkers directly. A frame problem is built as a Pondus model and solved by the Pondus analysis, so the member splitting, the load conversion and the reading of results are part of what is tested. A steel member is checked by the Pondus function that the design loop calls, `steel::check` for AISC 360 and `codes::check_ec3` for EN 1993-1-1, with the section properties the example prints converted to SI units. Where Pondus computes a property itself, such as C_b from the moment diagram or U from the end connection, the computed property is compared with the printed one as a separate row, so that an error in a property cannot hide inside an agreeing resistance. Where a design function bundles several clauses, the clause-level functions were exposed without changing any design result, so that the shear terms, the cracked and effective inertia, the torsion terms and the column section points can each be compared with the value the example prints.

3.3 Units and tolerances

US examples were converted exactly (1 in = 25.4 mm, 1 ksi = 6.894757 MPa, 1 kip = 4.448222 kN) and the results were converted back to the example's units before comparison. The default tolerance is 0.5 %, which is half a unit in the third significant figure that published examples print. A wider band was allowed only for three documented reasons. Some sources print fewer figures, such as a buckling reduction factor of 0.65. The plate rows are a mesh convergence study whose coarse meshes are expected to differ. The SI form of ACI 318, which Pondus implements, rounds several constants of the US form, so $0.17\sqrt{f'_c}$ MPa replaces $2\sqrt{f'_c}$ psi and gives a concrete shear strength 2.4 % higher, and each such row states the offset its constant implies.

3.4 Recording and treatment of discrepancies

The comparisons are Rust integration tests (`core/tests/calib_analysis.rs`, `calib_steel.rs`, `calib_concrete.rs`, `calib_walls.rs` and `calib_bridge.rs`) that assert each row against its tolerance and, when an environment variable names a directory, append the row to a CSV file. The tables and the figure in this paper are generated from those files, so the paper cannot quote a number that the tests did not produce. When a comparison failed, the cause was traced before anything was changed. A defect in Pondus was fixed, the whole test suite was run again, and the values before and after were recorded. A deliberate simplification of Pondus was kept, documented and, where it is conservative, asserted as such. A failure caused by the test model itself, such as a plate too thick for Kirchhoff tables, was corrected in the test and recorded as such.

4 Results overview

Table 1 summarises the 470 comparisons by aspect, and Figure 1 shows the distribution of the absolute errors. Taken together, 358 comparisons agree within 0.1 %, 430 within 0.5 % and 449 within 1 %. The median absolute error is below 0.001 % because many comparisons, such as the 107 slab coefficients, the statically determinate frames and the Wood–Armer moments, are reproduced exactly.

Table 1: Comparisons by suite and aspect. The pass column counts rows within their stated tolerance, and the error columns give the absolute relative error against the reference.

Suite	Aspect	Cases	Comparisons	Passed	Median abs. error (%)	Max abs. error (%)
analysis	cables	1	2	2	0.007	0.013
analysis	foundations	1	2	2	0.052	0.089
analysis	linear statics	9	19	19	0.000	0.255
analysis	modal	2	6	6	0.021	0.061
analysis	moving loads	3	3	3	0.000	0.000
analysis	nonlinear members	2	4	4	0.000	0.000
analysis	plastic collapse	5	5	5	0.000	0.000
analysis	plates	9	23	23	0.826	17.054
analysis	section properties	13	24	24	0.102	0.386
analysis	seismic	3	8	8	0.000	0.155
analysis	slab load transfer	2	7	7	0.000	0.000
analysis	stability	6	6	6	0.002	0.051
analysis	staged construction	1	2	2	0.002	0.003
analysis	supports and loads	5	6	6	0.000	0.000
analysis	time history	1	2	2	0.013	0.015
bridge	bridge moving load	1	40	40	0.045	0.646
concrete	RC beam deflection	3	5	5	0.013	0.452
concrete	RC beam flexure	2	5	5	0.079	1.584
concrete	RC beam shear	1	4	4	1.939	6.423
concrete	RC column P-M	6	17	17	0.099	0.739
concrete	RC column biaxial	1	2	2	8.311	8.396
concrete	RC column slenderness	2	5	5	0.000	0.846
concrete	RC footing	1	3	3	0.024	0.026
concrete	RC torsion	1	12	12	0.017	4.023
concrete	Wood-Armer	7	28	28	0.000	0.000

Suite	Aspect	Cases	Comparisons	Passed	Median abs. error (%)	Max abs. error (%)
concrete	masonry TMS 402	1	1	1	0.000	0.000
concrete	one-way slab	2	5	5	0.112	0.160
concrete	timber NDS	1	2	2	0.085	0.095
concrete	two-way slab coefficients	107	107	107	0.000	0.000
concrete	two-way slab design	2	9	9	0.018	0.062
steel	AISC 360 compression	3	6	6	0.047	0.152
steel	AISC 360 flexure	6	22	22	0.069	0.601
steel	AISC 360 interaction	1	2	2	0.119	0.169
steel	AISC 360 shear	1	1	1	0.088	0.088
steel	AISC 360 tension	2	5	5	0.072	0.240
steel	Connections	1	3	3	0.000	0.000
steel	EN 1992-1-1 (cross-check)	3	5	5	0.050	0.138
steel	EN 1993-1-1	6	28	28	0.010	0.221
walls	torsional coupling	1	5	5	0.012	0.043
walls	wall RSA	1	2	2	0.009	0.018
walls	wall modal	1	6	6	0.381	1.028
walls	wall time history	1	2	2	0.187	0.187
walls	wall-frame modal	6	19	19	0.350	1.124
All		**229**	**470**	**470**	**0.002**	**17.054**

Twenty-one comparisons differ from their reference by more than 1 %, and they fall into four groups whose causes are understood. Nine are the coarse meshes of the plate convergence study, which converge to within 0.9 % on the finest mesh (Section 5.2). Two are the biaxial column capacities by Bresler's load contour, which lie 8.2 % and 8.4 % below the exact skewed-axis capacity, so the method Pondus uses is conservative by that margin (Section 7.3). Six are concrete quantities that depend on the rounded SI constants of ACI 318, such as the 2.4 % in ϕV_c and the 6.4 % that this offset becomes in the required stirrup strength $V_s = V_u/\phi - V_c$, because the latter is a difference of two similar numbers. Four are the third frequencies of the walls, 1.03 % to 1.12 % low because Pondus's piers deform in shear, which the Euler–Bernoulli references leave out and which a Timoshenko solution reproduces (Section 5.4). None of the twenty-one is a defect of Pondus.

5 Analysis

5.1 Statics, supports and slab load transfer

The linear static cases reproduce their references to rounding. The ANSYS VM2 overhanging beam gives a bending stress of 11 404 psi and a mid-span deflection of 0.18246 in against the printed 11 400 psi and 0.182 in, which are themselves rounded values of Mc/I and $Ml^2/8EI$. VM4 gives the bar stress and the apex deflection exactly. A propped cantilever, a fixed-ended beam, a two-span continuous beam, a Gerber beam with an internal hinge, a Pratt truss and a three-dimensional L-frame loaded out of plane agree exactly with Roark, Hibbeler, Ghali and Neville, Kassimali and McGuire, Gallagher and Ziemian [11, 20–23]. A fixed-base portal under lateral load sways 0.012 % more than the slope-deflection formula because the formula ignores the axial strain of the columns. Springs, settlements, temperature changes and a rigid diaphragm reproduce their closed forms exactly. A two-way slab panel of 4 m × 6 m under 10 kPa sends 80 kN to each long beam and 40 kN to each short beam, as the 45° tributary lines require, and the beams' moments match $w_0(3L^2 - 4a^2)/24$ for the trapezoid and $w_0 l_x^2/12$ for the triangle.

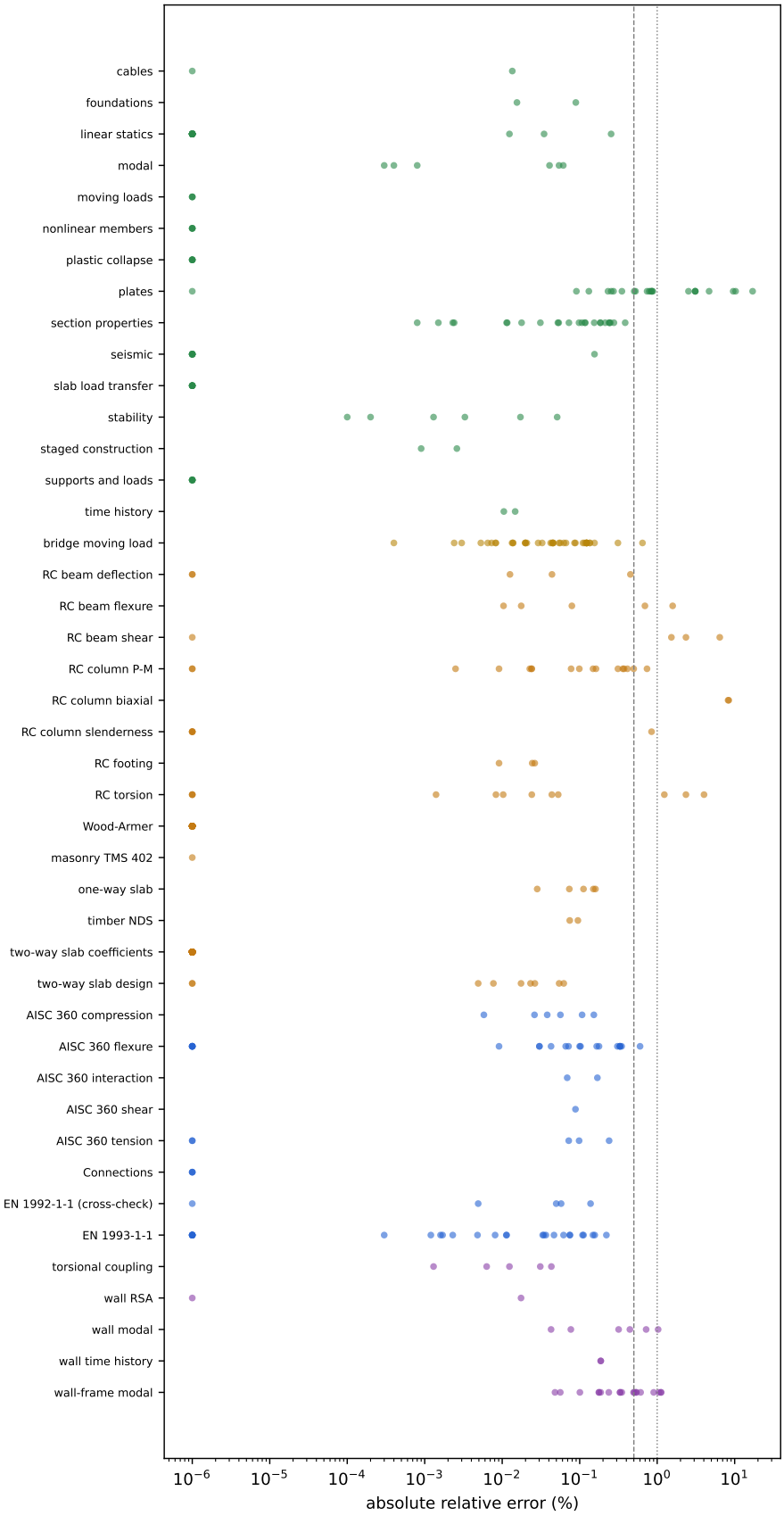


Figure 1: Absolute relative error of every comparison, by aspect, on a logarithmic scale (exact agreements are drawn at the left edge). The dashed and dotted lines mark 0.5 % and 1 %.

5.2 Plates

Table 2 reports the convergence of Pondus's plate slabs, meshed with MITC4 elements, towards the Kirchhoff solutions of Timoshenko and Woinowsky-Krieger [10] for square plates of side $a = 4$ m and thickness $a/100$ with $\nu = 0.3$. The edges are rigid beams that are free in torsion for the simply supported plate and rigid in torsion for the clamped plate. Moments converge faster than deflections, and on the finest mesh every quantity lies within 0.9 % of the table. A first attempt with $a/h = 20$ overshot the clamped deflection by 3.8 % because the Mindlin elements carry transverse shear deformation, which is physical and of order $(h/a)^2$, so the plates were made thin for the comparison. At the default slab mesh of 0.5 m, which on a 4 m panel is close to the $n = 9$ row, deflections run 3–5 % low while moments are within 1 %. Plate deflections used for serviceability should therefore be checked on a mesh of $a/16$ or finer.

Table 2: Error of Pondus's plate slabs against Timoshenko and Woinowsky-Krieger (Tables 8 and 35): $w = 0.00406\,qa^4/D$ and $M_x = 0.0479\,qa^2$ for the simply supported plate, and $w = 0.00126\,qa^4/D$, $M_x = 0.0231\,qa^2$ at the centre and $-0.0513\,qa^2$ at the mid-edge for the clamped plate. The mid-edge moment is extrapolated linearly from the two elements next to the edge. A rectangular plate with $b/a = 2$ at $n = 17$ gives −0.85 %, −0.27 % and −0.53 % for w, M_x and M_y.

Elements across a	SS w_{max}	SS M_x centre	Clamped w_{max}	Clamped M_x centre	Clamped M_x mid-edge
5	−10.29 %	−2.54 %	−17.05 %	+3.09 %	
9	−3.09 %	−0.83 %	−4.68 %	+0.35 %	+9.58 %
17	−0.79 %	−0.26 %	−0.85 %	−0.51 %	+3.11 %
33	−0.13 %	−0.09 %	+0.23 %	−0.74 %	+0.88 %

5.3 Modal, seismic and time-history analysis

A uniform cantilever of 32 elements gives its first three lateral frequencies within 0.001 % of the $\beta_n L$ values tabulated by Clough and Penzien [25], and a two-storey shear frame gives both frequencies within 0.06 % and the first-mode effective mass ratio of 0.9472 within 0.04 % of the closed form in Chopra [24], the small residual being the axial flexibility of the columns. A flat spectrum on that frame gives a CQC base shear within 0.16 % of the hand combination. The equivalent lateral force procedure of ASCE 7-22 [38] was checked on two frames, one in the velocity branch of the spectrum and one stiff enough that its period of 0.038 s lies below T_0, and the period, C_s and the base shear are exact after the defect described in Section 8 was fixed. No freely available printed ELF example with complete input data was obtained, so these rows check the equations rather than a printed example. An undamped single-degree-of-freedom system under a step ground acceleration reaches the dynamic amplification of 2 within 0.015 % at a time step of $T_n/200$.

5.4 Shear walls under earthquake and dynamic loading

Pondus models a shear wall by the wide-column analogy [39]: a pier on the wall's centre line carries the wall section $t \times l_w$ with a shear area of $5A/6$, and rigid arms join it to the wall's ends at every level, where beams and the floor diaphragm connect. Every wall case in this section was built with the same `add_walls` command an agent calls, so the wall generation, the seismic mass, the modal and CQC analysis and the time-history solver are tested together. Table 3 summarises the 34 comparisons, all of which pass after the fix of defect 13, and Appendix A.4 lists them.

A cantilever wall 0.3 m thick, 2 m long and 60 m high, divided into 20 storeys and carrying only its own weight, gives its first three frequencies 0.077 %, 0.44 % and 1.03 % below the $\beta_n L$ values of Clough and Penzien [25] and its effective modal masses within 0.72 % of the flexural cantilever [24]. The frequencies are low because the table describes an Euler–Bernoulli beam whereas Pondus's pier also deforms in shear, and a Timoshenko solution of the same wall predicts reductions of 0.062 %, 0.43 % and 1.01 %, so the shear area

accounts for the offset almost entirely. The same wall under an undamped step of ground acceleration of 0.1 g, integrated at $T_1/400$, reaches a peak top displacement of 1027.9 mm against 1026.0 mm by the superposition of six flexural modes, a dynamic amplification of 2.029 against 2.025, which exceeds 2 because the higher modes add to the first.

The interaction of a wall with moment frames, which governs the earthquake response of most wall buildings, was checked against the flexural–shear continuum of Heidebrecht and Stafford Smith [40] and Miranda and Taghavi [41]. In that model a flexural cantilever of rigidity EI is tied by rigid floors to a shear cantilever of rigidity GA, and the single parameter $\alpha H = H\sqrt{GA/EI}$ moves the structure from a pure wall at $\alpha H = 0$ to a pure frame as αH grows. The test solves the continuum's frequency equation itself, after checking the solver against both limits and against an independent beam finite-element solution that agreed to four digits. The Pondus model is the 60 m wall joined by rigid diaphragms at 40 levels to two frames whose columns, sized to give the required GA, are axially rigid, so the frames act in shear alone as the continuum assumes. Over αH from 0.5 to 8, the first frequency lies 0.18 % to 0.33 % below the continuum, the second and third 0.50 % to 1.12 % below, and the first-mode effective mass 0.05 % to 0.18 % above. The wall's shear deformation explains about half of the first-mode offset at $\alpha H = 8$ and most of it at low αH. The rest, about 0.1 %, comes from the minimum density of 1 kg/m^3 that Pondus gives weightless members, which adds about 0.2 % to the mass of these lightly loaded models. At $\alpha H = 8$ the first-frequency error falls from 0.61 % at 10 storeys to 0.35 % at 20 and 0.33 % at 40, converging on that floor.

Torsional coupling, the mechanism that makes a wall on one side of a plan dangerous in an earthquake, was checked on a one-storey system of the kind Chopra [24] treats in closed form. A 10 m square floor carries its mass on three edges, so the mass centre lies at $x = 6.67$ m, and it rests on four corner columns pinned at the top and on a 4 m wall along the fourth edge. The three coupled modes agree with the 3×3 eigenproblem of the rigid floor within 0.031 %, the share of the mass in y of the two coupled modes within 0.043 %, and the CQC base shear in y under a flat spectrum of 0.4 g within 0.017 %. Reaching that agreement needed two corrections to the test model, both recorded in the notes: the floor beams were first hinged at both ends, which made a mechanism with the pinned column tops, and then they were too flexible axially and in plan for the 47 Hz torsional mode, because only the model's own nodes join the diaphragm while the mass lies along the beams between them. The static stiffness of the floor matched the reference to six digits throughout, so neither correction concerned Pondus.

Table 3: Comparisons for shear walls under earthquake and dynamic loading. The negative frequency errors of the first two rows are the shear deformation of the wall, which the Euler–Bernoulli references leave out.

Case	Reference	Quantity	Error
cantilever wall, 20 storeys	$\beta_n L$ [25]; flexural effective masses	f_1, f_2, f_3; mass of modes 1–3	−0.08, −0.44, −1.03 %; +0.04 to +0.72 %
wall + shear frames, αH = 0.5, 1, 3, 8, 40 storeys	flexural–shear continuum [40, 41]	f_1; f_2, f_3; mass of mode 1	−0.18 to −0.33 %; −0.50 to −1.12 %; +0.05 to +0.18 %
same, αH = 8, 10 / 20 / 40 storeys	continuum	f_1	−0.61, −0.35, −0.33 %
one storey, wall on one edge	3×3 rigid-floor eigenproblem [24]; CQC [12]	f_1–f_3; mass in y; V_{RSA}; W	−0.001 to −0.031 %; −0.043, +0.006 %; +0.017 %; exact
cantilever wall, step of 0.1 g	superposition of six modes [24]	peak top displacement; amplification	+0.19 %; +0.19 %

The wall-frame cases exposed defect 13 (Section 8). The rigid arms had an absolute stiffness, $E = 2.5 \times 10^7$ MPa on a section with $A = I = J = 1$, regardless of the wall they belonged to. A slender wall coupled to soft frames then puts a stiffness contrast of order 10^9 into the stiffness matrix on top of the cantilever's own poor conditioning, and the solution loses precision silently, with global equilibrium still satisfied and no warning raised. At $\alpha H = 0.5$ and 40 storeys the first frequency came out 11.5 % high and the static top deflection 21 % low, whereas the same model at 10 or 20 storeys, or without its diaphragms, was right. The arms now take their own wall's section and 1000 times its modulus, so they remain rigid against the wall without an absolute stiffness, and models saved earlier have their arms rebuilt when they are opened.

5.5 A full-scale wall building on a shake table

The comparisons so far are calibration in the sense of Section 3, because their references are exact for the model Pondus solves. A user also needs to know how close that model comes to a real building, which is validation in the vocabulary of ASME V&V 10 [6]. Pondus was therefore compared with the measured dynamic properties of the full-scale seven-storey reinforced concrete wall building slice tested on the UCSD–NEES shake table between October 2005 and January 2006 [42, 43]. These comparisons are recorded separately (`core/tests/validation_ucsd.rs`) and are not counted in the 470 calibration rows. Their differences combine the program's error with the modeller's choices and with the physics that a linear model leaves out. The slice resists lateral load in the direction of shaking with a single rectangular web wall 3.658 m long and 19.20 m high in seven storeys of 2.743 m, 0.203 m thick in the first and seventh storeys and 0.152 m between. It carries 3.66×8.12 m slabs on pin-ended gravity columns, and a flange wall and a post-tensioned precast pier provide transverse and torsional stability [44]. The measured concrete strength averaged 41.4 MPa against 27.6 MPa specified [44].

The Pondus model is the web wall alone, built with `add_walls` in three segments, on a fixed base, with $E = 4700\sqrt{41.4} = 30.2$ GPa and the gross section. The seismic weight was taken as $W = V_b/C_s = 292/0.15 = 1947$ kN from the design base shear and coefficient that the designers report [43]. The wall's own weight of 281 kN comes from its concrete, and the remaining 1666 kN was placed at the floors in proportion to the slab thickness. The flange wall, the post-tensioned pier, the frame action of the slabs with the gravity columns and the flexibility of the shake-table foundation were left out, as a designer would leave them out of a model of the lateral system. Figure 2a shows the model as Pondus draws it. Table 4 compares the results with the frequencies identified from the test by Moaveni et al. [45] and Panagiotou et al. [43] and with two published models: the designers' uncracked three-dimensional model and the nonlinear ANSR-II model of the blind-prediction study [44].

Table 4: The UCSD–NEES seven-storey building slice: Pondus against the measured periods and frequencies and against two published models. Measured values are from white-noise and ambient-vibration tests, model values are the periods the cited authors computed, and every reference value is as printed.

Pondus model	Quantity	Reference	Value	Pondus	Diff. (%)
gross section, E = 4700 √(41.4) MPa	f1, ambient vibration before testing (uncracked) (Hz)	Moaveni et al. (2011), measured	2.23	1.846	-17.2
	T1, first 0.03 g RMS white-noise test (s)	Panagiotou et al. (2010), measured	0.51	0.5419	+6.2
	T1, designers' uncracked 3-D model (s)	Panagiotou et al. (2010), model	0.5	0.5419	+8.4
	T1, blind-prediction model (ANSR-II) (s)	Kelly (2007), model	0.557	0.5419	-2.7
	f2, ambient vibration at state S0 (Hz)	Moaveni et al. (2011), measured	10.53	10.62	+0.9

Pondus model	Quantity	Reference	Value	Pondus	Diff. (%)
	T2, low-amplitude tests (s)	Panagiotou et al. (2010), measured	0.1	0.09414	-5.9
	T2, blind-prediction model (ANSR-II) (s)	Kelly (2007), model	0.094	0.09414	+0.2
	f3, ambient vibration at state S0 (Hz)	Moaveni et al. (2011), measured	24.58	26.1	+6.2
	effective mass, mode 1	Kelly (2007), model	0.663	0.6255	-5.7
	effective mass, mode 2	Kelly (2007), model	0.175	0.2128	+21.6
gross section, E = 4700 $\sqrt{}$(27.6) MPa (specified f'c)	T1, first 0.03 g RMS white-noise test (s)	Panagiotou et al. (2010), measured	0.51	0.5997	+17.6
	f1, ambient vibration before testing (uncracked) (Hz)	Moaveni et al. (2011), measured	2.23	1.668	-25.2
	f2, ambient vibration at state S0 (Hz)	Moaveni et al. (2011), measured	10.53	9.598	-8.8
0.70 Ig (ACI 318-19 6.6.3.1.1, uncracked wall), E from f'c 41.4 MPa	T1, first 0.03 g RMS white-noise test (s)	Panagiotou et al. (2010), measured	0.51	0.6476	+27.0
	f1, ambient vibration before testing (uncracked) (Hz)	Moaveni et al. (2011), measured	2.23	1.544	-30.8
	f2, ambient vibration at state S0 (Hz)	Moaveni et al. (2011), measured	10.53	8.887	-15.6
	T1 after test EQ1 (moderate, cracking) (s)	Panagiotou et al. (2010), measured	0.65	0.6476	-0.4
0.35 Ig (ACI 318-19 6.6.3.1.1, cracked wall), E from f'c 41.4 MPa	T1, first 0.03 g RMS white-noise test (s)	Panagiotou et al. (2010), measured	0.51	0.9159	+79.6
	f1, ambient vibration before testing (uncracked) (Hz)	Moaveni et al. (2011), measured	2.23	1.092	-51.0
	f2, ambient vibration at state S0 (Hz)	Moaveni et al. (2011), measured	10.53	6.284	-40.3
	T1 after test EQ2 (s)	Panagiotou et al. (2010), measured	0.82	0.9159	+11.7
	T1 after test EQ3 (s)	Panagiotou et al. (2010), measured	0.88	0.9159	+4.1
	T1 after test EQ4 (Sylmar, near-fault) (s)	Panagiotou et al. (2010), measured	1.16	0.9159	-21.0

With gross sections, Pondus gives a first period of 0.542 s, 6.2 % longer than the 0.51 s identified from the first low-amplitude white-noise test, 8.4 % longer than the designers' uncracked three-dimensional model and 2.7 % shorter than the blind-prediction model. The mode shapes have the expected cantilever form (Figure 2b), and the second mode agrees more closely, at 10.62 Hz against 10.53 Hz identified from ambient vibration (+0.9 %) and 0.0941 s against 0.094 s in the blind-prediction model. The first mode is the one the omitted elements stiffen, because the flange wall, the post-tensioned pier and the slab–column frames act through the overturning of the whole slice, which Kelly [44] and Panagiotou et al. [42] found to be significant, so a bare-wall model is expected to be somewhat long in T_1 while its second mode, dominated by the wall's own curvature, stays close. The ambient-vibration frequency of 2.23 Hz measured before any testing is 17 % above Pondus, because at those micro-amplitudes the concrete was uncracked with full tension stiffening,

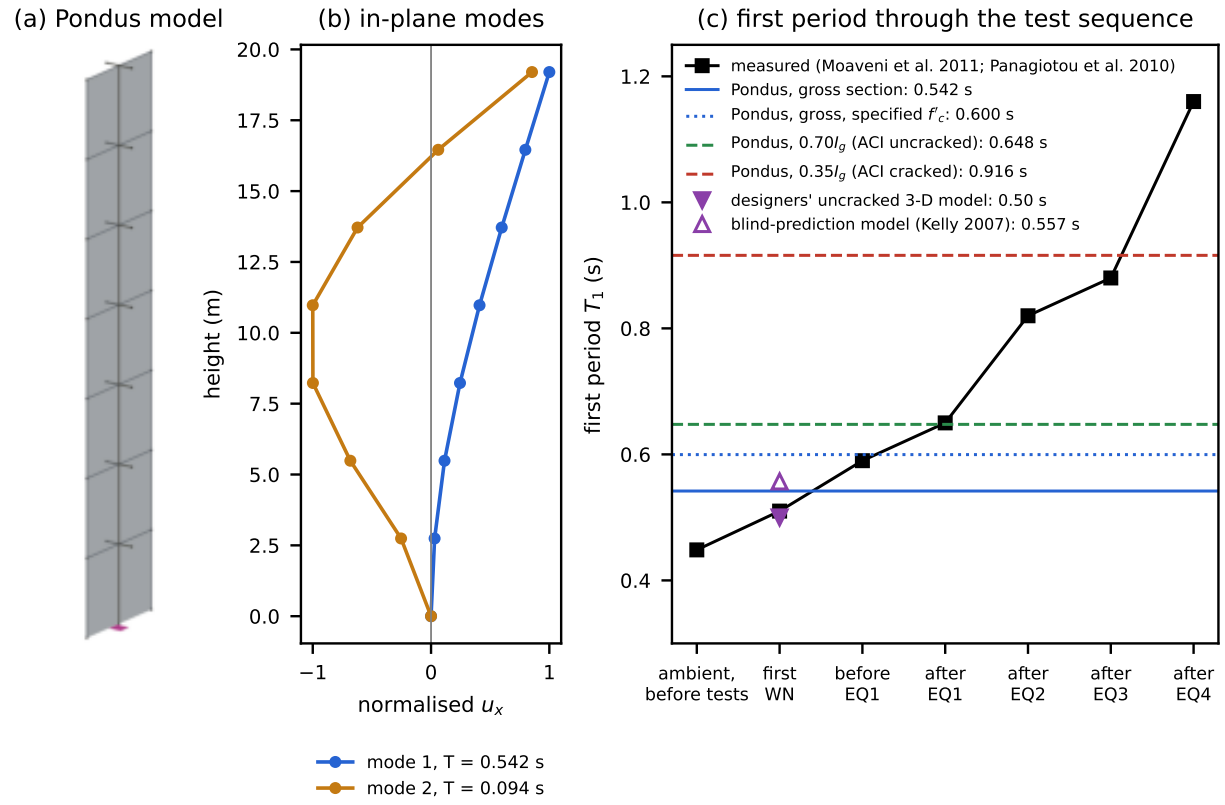


Figure 2: The UCSD–NEES seven-storey building slice in Pondus. (a) The model as Pondus draws it: the web wall in three thickness segments, its piers and rigid arms, and the weightless stubs that carry the floor masses. (b) The first two in-plane mode shapes along the wall's centre line. (c) The first period identified at each stage of the test [43, 45] against Pondus with four stiffness assumptions and the two published models.

which twenty-five white-noise tests then reduced to the state behind the 0.51 s [43]. The mass is a second source of uncertainty, because the reported C_s of 0.15 has two figures, which leaves W uncertain by about ±3 % and T_1 by ±1.7 %.

The stiffness that an engineer assumes matters far more than these differences, as Figure 2c shows against the periods identified through the test sequence. With E from the specified rather than the measured strength the first period lengthens to 0.600 s (+17.6 %). The effective stiffnesses of ACI 318-19 6.6.3.1.1 bracket the damaged states instead: $0.70I_g$ for uncracked walls gives 0.648 s, which matches the 0.65 s identified after the first earthquake test EQ1 within 0.4 %, and $0.35I_g$ for cracked walls gives 0.916 s, between the 0.82 s and 0.88 s identified after EQ2 and EQ3 and well short of the 1.16 s after the near-fault Sylmar record EQ4, which yielded the base of the wall [43]. The close match after EQ1 is a single observation and should not be read as a calibration of the code factor. The comparison therefore places the program's own error, below 1 % in Sections 5.1 to 5.4, well inside the 3–8 % that modelling decisions add for a real building and the 30–80 % range that cracking adds, which is the order of priorities a user of Pondus should keep in mind.

5.6 A continuous highway bridge under HL-93 loading

The moving-load analysis was calibrated against the published design of a two-span continuous composite steel I-girder bridge, Design Example 2A of the FHWA Steel Bridge Design Handbook [46], whose two 90 ft spans carry four plate girders designed to AASHTO LRFD [47]. Its Table 6 prints the unfactored live load moments per lane at the tenth points of span 1 under the HL-93 loading. These are the envelopes of the design truck, with its rear axle spacing free between 14 and 30 ft, and of the design tandem, the same envelopes with 33 % dynamic allowance plus the 0.64 klf lane load, and the special negative-moment loading of 90 % of two trucks at least 50 ft apart plus 90 % of the lane. The example analyses the girder with short-term composite moments of inertia of 48,806 in.^{4} in the positive bending region and 50,027 in.^{4} within 27 ft of the pier, and Pondus used the same girder, divided at the tenth points. The vehicle envelopes come from Pondus's moving-load analysis at a step of 0.5 ft over nineteen trains, the truck at nine rear spacings in both directions and the tandem. The lane load was placed on the parts of Pondus's influence line with the sign sought, because AASHTO allows it to be discontinuous.

All 40 comparisons agree within their tolerance, with a median error of 0.045 % and a largest of 0.65 % (Figure 3). The vehicle envelopes agree within 0.16 % at every tenth point, and the special loading gives −2020.1 kip-ft at the pier against the printed −2020 kip-ft. At 0.9L the influence line changes sign inside span 1, and its positive part carries 31.5 kip-ft of lane load both in Pondus and in an independent integration, whereas the table implies 29.6 kip-ft, which points to a coarser placement of the lane in the example, so that row was allowed 1 %. Two parts of the table were not compared. The special loading is defined only between the points of dead-load contraflexure, and the values printed away from the pier match no placement of two trucks. At 0.8L the printed −967 kip-ft would need a two-truck effect of −611 kip-ft, whereas neither Pondus nor an independent search over every placement finds more than −479.5 kip-ft, so the value appears to be an error in the table.

The first run exposed defect 14. Pondus read the envelope at a station from the engine's extreme over each element touching it, but the engine takes that extreme over both ends of an element, so a station carried the value one step away. Where the envelope changes quickly the result was too high, 958.7 instead of 951 kip-ft at 0.6L (+0.81 %), and an independent influence-line solution confirmed the table. The engine now also keeps the envelope at each end of an element, and Pondus reads the value at the station itself, interpolates inside an element and gives an extreme that falls between two stations to the nearer one. The absolute maximum moment of a simple span under a moving pair of axles is therefore still found exactly (Section 5.7).

5.7 Stability, plastic collapse and the other advanced analyses

Euler columns with pinned, cantilever, fixed-pinned and fixed-fixed ends give their critical loads within 0.05 % of Timoshenko and Gere [18], including the printed coefficient 20.19 for the fixed-pinned case. A fixed-

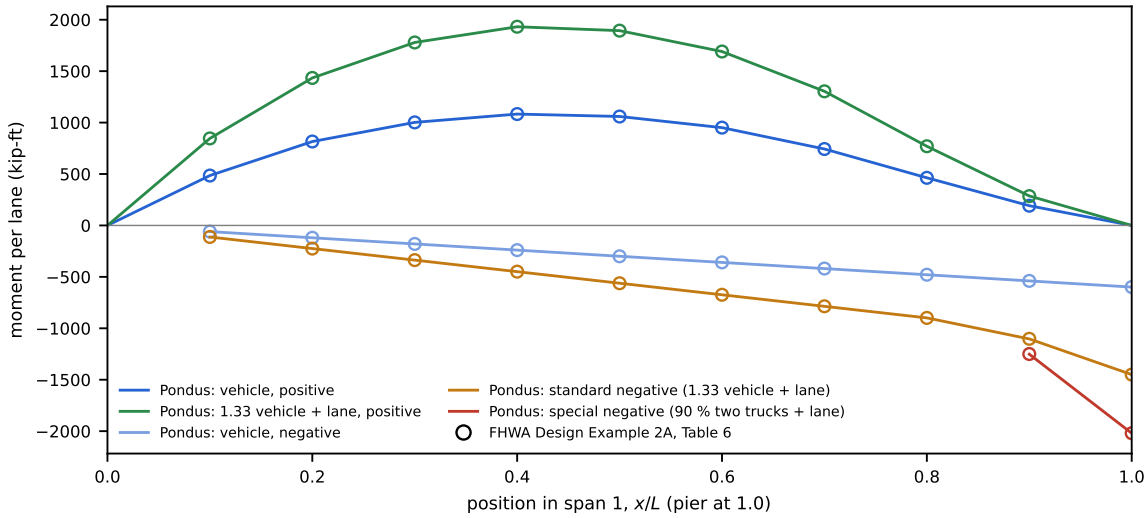


Figure 3: Live load moments per lane in span 1 of the FHWA two-span bridge under HL-93 loading: Pondus (lines) against Table 6 of the design example (circles). The special loading is drawn where it is defined and compared.

base portal with equal beam and column stiffness buckles in sway within 0.001 % of the root of $x \cot x = -6/G$, the alignment-chart equation, when its members are axially rigid, and 0.36 % lower when their real axial stiffness is kept. The P-Delta amplification of a cantilever at 0.4 P_{cr} matches $3(\tan u - u)/u^3$ within 0.0001 %. Five plastic collapse cases of Neal [26] and Horne [27], from a simple beam to a fixed-base portal whose combined mechanism governs, are reproduced exactly by the pushover analysis, and the two-span case shows that the hinge forms at $0.414L$ inside the span rather than at an element end. A moving pair of axles gives the absolute maximum moment $P(L - s/2)^2/2L = 405$ kN·m exactly, influence lines peak where Hibbeler [21] says they should, a cantilever propped after its dead load carries the staged moments within 0.003 %, a cable stay elongates within 0.014 % of Ernst's equivalent modulus [29], and a beam on a Winkler foundation deflects within 0.015 % and bends within 0.09 % of Hetényi [28]. The tension-only brace and the lift-off support reproduce their statics exactly.

5.8 Section properties

The torsion constants of rectangles with b/t from 1 to 10 from Pondus's section finite elements lie within 0.39 % of the coefficients that Timoshenko and Goodier [19] print to three figures, and the frame formula Pondus uses for rectangular members lies within 0.19 %. The warping constant of an I section and the shear centre of a thin channel are within 0.07 % of their closed forms, and the area and second moment of five rolled H shapes computed from their plates and fillets agree with JIS G 3192 [48] within 0.02 % and 0.25 %.

6 Steel and Eurocode design

Table 5 lists the governing quantities of the steel examples, and the full set of 72 rows, including the intermediate values, is in the appendix. Every AISC 360 resistance agrees with the AISC Design Examples within 0.35 %, and the largest deviation of the suite, −0.60 %, is the nominal moment at the printed L_r of example H.3, where the Manual's tabulated r_{ts}, h_o and C_w are not mutually exact. Pondus computes C_b from the moment diagram within 0.35 % of the printed values of F.1-2B and F.1-3B. The EN 1993-1-1 classification, buckling reduction factors, elastic critical moment under a top-flange load and Annex B interaction factors agree with the SCI/CTICM and P364 examples within 0.22 %. The EN 1992-1-1 cross-check of concrete beams reproduces the JRC flexural and shear resistances within 0.14 %.

Table 5: Governing quantities of the steel and Eurocode examples [9, 13–15].

Example	Quantity	Printed	Pondus	Error
AISC D.1, W8×21	$\phi_t P_n$, yielding (kips)	277	277.2	+0.07 %
AISC D.1, W8×21	A_n (in^2), U	4.76, 0.908	4.760, 0.907	0.00 %, −0.10 %
AISC D.1, W8×21	$\phi_t P_n$, rupture (kips)	211	210.5	−0.24 %
AISC E.1C, W14×132	$\phi_c P_n$ (kips)	892	893.4	+0.15 %
AISC E.1D, W14×90	$\phi_c P_n$ (kips)	927	927.5	+0.06 %
AISC F.1-2B, W18×50	$\phi_b M_n$ (kip-ft)	304	305.0	+0.33 %
AISC F.1-3B, W18×50	$\phi_b M_n$ (kip-ft)	288	287.9	−0.03 %
AISC F.3B, W21×48	$\phi_b M_n$, flange local buckling (kip-ft)	398	398.0	−0.01 %
AISC G.1B, W24×62	$\phi_v V_n$ (kips)	306	305.7	−0.09 %
AISC H.3, W14×82	H1-1b	0.873	0.874	+0.17 %
SBE A.1, IPE 330	M_{cr}, top-flange load (kN·m)	113.9	113.86	−0.04 %
SBE A.5, HE 300 B	χ_z	0.671	0.6705	−0.07 %
P364 Ex. 9, UKC 356×368×129	$N_{b,z,Rd}$ (kN)	3678	3672	−0.16 %
SBE-MS8 Ex. 2, IPE 450	class under N + M, χ_z	1, 0.865	1, 0.8660	0, +0.11 %
JRC EC2 §3.2.1.4	$V_{Rd,s}$ (kN)	115.52	115.36	−0.14 %

The tension rupture rows deserve a comment because they changed during the calibration. Pondus first took the effective net area equal to the gross area, so it reproduced the yielding capacity of D.1 but reported a rupture capacity of 300 kips where the example's governing value is 211 kips. It now describes the end connection of a tension member by its kind, the connected elements, the bolt diameter, the holes in each flange and in the web, the bolts per line and the connection length. From these it computes A_n with holes of width $d + 1/8$ in and the shear lag factor U as the larger of case 2 ($1 - \bar{x}/l$, with $\bar{x}$ of the split tee) and case 7 of AISC Table D3.1. With the connection of example D.1 it gives $A_n = 4.760$ in^2 and $U = 0.907$ against the printed 4.76 in^2 and 0.908, the small difference in U coming from the fillets that the tabulated $\bar{x} = 0.831$ in includes. The same description gives the EN 1993-1-1 net section resistance $0.9 A_{net} f_u / \gamma_{M2}$.

7 Concrete, timber and masonry design

7.1 Slab coefficients

The two-way slab design of Pondus follows Method 2 of ACI 318-63 [35], the method that the Thai standard EIT 1008 adopts, and the coefficient table had been written from a secondary reproduction with only its interior-panel row confirmed. The calibration found a scan of the original Table I in lecture notes of the University of Basrah [34], and every one of its 91 coefficients, for the five edge conditions, six span ratios and the long span, equals Pondus's table exactly. A further 13 rows confirm that span ratios below 0.5 take the column for 0.5 and less. Both worked examples of the notes are reproduced through Pondus's slab design within 0.06 %. The notes themselves swap two row labels and transpose one coefficient in an example, which the comparison exposed and which the calibration records. The one-way coefficients of ACI 318-19 Table 6.5.2 agree within 0.16 % with a StructurePoint reproduction of Wight [32] Example 5-7, except where Pondus is deliberately or newly conservative, as described in Section 8.

7.2 Beams

Flexure, shear, torsion and deflection were compared clause by clause. The INTI singly reinforced beam gives ϕM_n within 0.01 %, and the required tension steel of a simply supported beam by Wight [32] agrees within 0.02 %. The torsion terms of Wight Example 7-2, a 14×24 in cantilever under 28 kip-ft, agree within 0.05 % for A_{oh}, p_h, A_o, ϕT_{th}, ϕT_{cr}, A_t/s, A_l, the spacing limit and the cross-section limit of 22.7.7.1. The quantities that contain the rounded SI constants, ϕV_c, V_s, $A_{l,min}$ and the minimum transverse steel, differ by the offsets those constants imply and by no more. For deflection, the cracked section and cracked inertia agree within 0.05 %, while the effective inertia is the expression of ACI 318-19 Table 24.2.3.5, which the 318-14 example replaces with Branson's equation, so the effective inertia was evaluated independently and agrees exactly, and the 7.6 % gap to the example's Branson value is the intended change between editions.

7.3 Columns and footings

The interaction diagram of a 16×16 in tied column with twelve #9 bars reproduces P_0 and $\phi P_{n,max}$ exactly and the axial forces and moments at the four printed neutral-axis depths within 0.74 %, the residue of $\beta_1 = 0.804$ in the SI expression against 0.80 in the psi expression. The search that Pondus's column design uses returns ϕM_n at each printed ϕP_n within 0.37 %. For biaxial bending Pondus uses Bresler's load contour [49] with $\alpha = 1.15$ on its own uniaxial capacities, and on the example of Pincheira and co-authors reproduced by StructurePoint it gives a capacity 8.2–8.4 % below the exact skewed-axis point, which quantifies the conservatism of the method rather than an error of the code. The non-sway moment magnifier, the critical load by option (a) of 6.6.4.4.4 and the minimum moment agree within 0.85 %. The isolated footing of Wight [32] Example 15-2 gives the moment at the column face and the reinforcement within 0.03 %.

7.4 Plate reinforcement, timber and masonry

The Wood–Armer moments [50, 51] were evaluated on seven moment fields that cover every branch of the normal and special rules, and all 28 design moments agree exactly. The NDS column stability factor of a timber column [37] agrees with the Dlubal example within 0.08 %, after the calibration noticed that the example's printed adjusted compressive strength does not follow from its own stability factor. No accessible published masonry example was obtained, so the TMS 402 flexural check [52] was compared with its closed form only.

8 Defects found by the calibration

Table 6 lists fourteen defects with their effect on a design before the fix. Twelve were found by the calibration, and two, the staged member forces and the sign of the Winkler coupling, by the closed-form tests that preceded it during development, which are included because they change design forces in the same way. Six were unconservative, four were conservative, three returned wrong results without a clear direction, and one, the engine's single coefficient of thermal expansion, overestimated restraint forces in concrete members by 20 %. Every defect was fixed and every fix was followed by a run of the whole test suite. The development tests also corrected two reporting errors that do not enter a design, a pushover capacity curve that plotted the displacement increment of each event as if it were the total and an influence line sampled beside its section, whose peak was 3.6 % low.

Table 6: Defects found by the calibration (1–10, 13, 14) and by the closed-form tests that preceded it (11, 12), with their effect before the fix. Unconservative effects are in bold. Defect 13 is counted as without a clear direction, because the overstiff model underestimates drift but overestimates the spectral acceleration of a shortened period.

No.	Defect	Where	Effect before the fix	After
1	ELF C_s read from the ascending branch of the spectrum below T_0	seismic	C_s 0.0738 instead of 0.125, base shear **41 % low**	exact
2	moving-load envelope taken from element ends only	engine, wrapped	absolute maximum moment 400 instead of 405 kN·m, **1.2 % low**	exact
3	Winkler moduli passed to the engine as $[k_y, k_z]$ instead of $[k_z, k_y]$	analysis wrapper	a beam on a vertical foundation had none: singular or 1.4×10^{11} mm	within 0.015 %
4	one effective length for both axes	AISC and EC3	E.1D $\phi_c P_n$ 597 instead of 927 kips, 36 % low	+0.06 %
5	EC3 class from the pure-compression web limit for any member ever compressed	EC3 wrapper	IPE 450 class 4 instead of class 1	as printed
6	EC3 C_2 left equal to C_1 for a top-flange load	EC3 wrapper	M_{cr} 78.6 instead of 113.9 kN·m, 31 % low	−0.04 %
7	EC2 cross-check took the stirrups at the main-bar grade	EC2 wrapper	$V_{Rd,s}$ **66 % high** for SR24 stirrups	exact ratio
8	tension rupture on the gross area	AISC	D.1 rupture 300 instead of 211 kips, **42 % high**	−0.24 %
9	first interior support of a one-way end span at $wl^2/10$, the value for three or more spans	slab coefficients	**10 % low** for two-span slabs	$wl^2/9$, safe for any number of spans
10	one coefficient of thermal expansion, 12×10^{-6}/°C, for every material	engine default	restraint forces in concrete 20 % high	per material
11	staged construction member forces from $K \cdot u$ without fixed-end forces	engine	end moment of a loaded beam −30.60 instead of −31.08 kN·m, **1.6 % low**	exact
12	sign of the w–θ_y coupling in the vertical Winkler foundation matrix	engine kernel	a uniformly loaded beam on a uniform foundation settled unevenly	uniform, 1.500 mm
13	rigid arms of shear walls with an absolute stiffness ($E = 2.5 \times 10^7$ MPa, $I = 1$ m^4), a contrast of order 10^9 in K	wall generation	40-storey wall-frame: f_1 11.5 % high, top deflection 21 % low, no warning	−0.18 %, +0.08 %
14	moving-load envelope at a point read from the engine's extreme over each element, which includes the element's far end	engine, wrapped	FHWA bridge: vehicle M^+ at 0.6L 958.7 instead of 951 kip-ft, +0.8 %	−0.013 %

The location of the defects is the main lesson. Only one of the fourteen, the sign in the Winkler foundation matrix, was an error of a numerical kernel. The frame stiffness matrices, the solvers, the eigenvalue analysis,

the plastic event-to-event algorithm and the code formulas of the engine's checkers were correct wherever they were reached. Nine defects sat in Pondus's own code that decides what the kernel is asked: which axis a modulus belongs to, which branch of a spectrum a period falls on, which length the minor axis buckles over, which web limit a beam-column is classed by, which coefficient a destabilising load takes, which strength a stirrup has, which area a bolted member ruptures across, which moment coefficient a two-span slab needs and how stiff a modelling device such as a rigid arm may be made before the solver loses precision. Defect 13 is instructive because the kernel was right and the inputs were legal: the error came from a modelling convenience, an arbitrarily large stiffness, that is harmless in small models and was invisible to the existing wall tests, which had 20 storeys or fewer. Four of the defects, the two in the moving-load envelope, the staged member forces and the thermal coefficient, are limitations of the vendored engine that its own suite did not expose because they only matter for the inputs a design program sends. That pattern agrees with the general experience of verification that interfaces and defaults, not algorithms, carry most of the risk [7], and it matters more when an agent is the user, because an agent has no habit of distrusting a number that looks plausible.

A second lesson concerns the tests that existed before the calibration. Pondus already had tests against closed-form results, and they passed. They did not catch defect 3 because they gave the Winkler modulus as $[k, k]$, they did not catch defect 1 because no test structure was stiff enough to fall below T_0, and they did not catch defect 8 because no test had a bolted tension member. Published examples do better than self-written tests because their authors chose cases for engineering reasons rather than to exercise the code, and because the printed intermediate values pin each clause separately.

9 Limitations of the calibration

The calibration has limits that a user of Pondus should know. The ELF vertical force distribution is not calibrated because Pondus does not compute it: it scales the response spectrum result to the ELF base shear instead. Only a single-degree-of-freedom time history was calibrated. The cables are small-displacement trusses with Ernst's modulus, so slack cables under large transverse loads are outside what Pondus solves. The plate results are Mindlin results, which exceed the Kirchhoff tables for thick slabs by a physical shear deformation and, at the default mesh, run 3–5 % low in deflection. In steel design, the AISC shear tab and the AISI cold-formed checks were not compared with printed examples because none was obtained in text form, and block shear, the tension increase of C_b in H1.2, the thickness reduction of the yield strength in EN 10025 and the lateral-torsional buckling method of 6.3.2.3 are not implemented. In concrete design, Pondus has no capacity function for a given doubly reinforced section, assumes a sustained load ratio of 0.6 in the slender column stiffness, and does not design punching shear or slab deflection, and the masonry interaction check is the engine's linear sum.

The shear-wall calibration has limits of its own. The walls are wide columns, so they carry no in-plane flexibility between the pier and the wall's ends, which is right for walls whose length is small against the storey span but not for long walls with openings or for coupled walls, whose coupling beams were not calibrated. The references are linear and elastic: cracked-section stiffness, the nonlinear response of walls and the ACI 318 design of boundary elements were not compared with published examples. The wall-frame comparisons rest on a continuum solved in the test rather than on printed numbers, which was mitigated by checking the solver against its two limits and against an independent finite-element solution. The validation rests on a single building, on its linear dynamic properties only and on a seismic weight inferred from the design base shear, and Pondus's linear time history was not compared with the measured earthquake response, which was nonlinear from the second test on. Weightless members receive a minimum density of 1 kg/m^3, which in these lightly loaded benchmark models adds about 0.2 % to the mass and is negligible in a building.

The sources also have limits. Some published values reach the calibration through reproductions, such as Stabileo's records of the AISC and Eurocode examples, StructurePoint's versions of textbook examples and a lecture-note scan of the 1963 coefficient table, so a transcription error in a reproduction would pass

into the reference. Where a reproduction contradicted itself, as in a misprinted timber strength, a swapped label in the slab notes and an inconsistent load table in a slenderness example, the calibration followed the calculation that the source actually performs and recorded the inconsistency. Most clauses are checked by one or two examples, which is enough to expose a wrong formula but not to prove that every branch of a long clause is right. The tolerance of 0.5 % was chosen to match the precision of printed examples, and it cannot detect an error smaller than that.

10 Conclusions

Pondus reproduces 430 of 470 published, closed-form and independent reference values within 0.5 % and 449 within 1 %, across analysis, shear walls under earthquake and dynamic loading, a continuous bridge under moving load, steel, concrete, timber and masonry, and the remaining 21 differences are explained by mesh convergence, a conservative method for biaxial bending, the rounded SI constants of ACI 318 and the shear deformation of walls that the Euler–Bernoulli references omit. Walls coupled to frames reproduce the flexural–shear continuum within 0.33 % in the first frequency over αH from 0.5 to 8, and a wall on one side of a plan reproduces the coupled torsional modes and the CQC base shear within 0.05 %. The HL-93 envelopes of a published two-span bridge design are reproduced within 0.16 % for the vehicles and within 0.65 % with the lane load. Against the full-scale UCSD–NEES wall building, a bare-wall Pondus model with gross sections lies 6.2 % long in the first period and 0.9 % off in the second-mode frequency, which is the size of the effect of the elements a lateral-system model leaves out, whereas the choice between gross and cracked stiffness changes the first period by 20 % to 70 %. The ACI 318-63 Method 2 coefficient table, previously confirmed for one row, now agrees with the published table in all 91 coefficients. The calibration and the closed-form tests before it found and fixed fourteen defects, six of them unconservative and three of them capable of changing a design by 40 % or more, and thirteen of the fourteen lay in the code between the model and the numerical kernels rather than in the kernels themselves.

For software that agents will drive, these results suggest a practical rule. Verification effort should follow the data across every interface, from the model to the inputs of each check, and should use published examples whose intermediate values pin each clause, because that is where errors are made and where neither the agent nor a self-written test is likely to notice them. The calibration suite is part of the program's test suite, so every future change to Pondus is checked against the same 470 references.

11 Declaration of AI use

The program and its test suite were coded with the help of the agentic coding tool Claude Code and the Claude Fable model (Anthropic). The author designed the calibration, chose the references, checked the results and takes full responsibility for the content of this paper.

12 Data availability

Pondus is free software under the GNU Affero General Public License, version 3 only, and its complete source, including the vendored Stabileo engine with its three marked changes, is available at https://github.com/Sompote/pondus. The calibration is reproduced by `cargo test --release -p pondus-core --test calib_analysis --test calib_steel --test calib_concrete --test calib_walls --test calib_bridge` with `PONDUS_CALIB` set to an output directory, and the validation by `--test validation_ucsd` with `PONDUS_VALID`, which write the CSV files from which `paper/make_tables.py` generates every table and figure of this paper. The notes in `calibration/NOTES_*.md` give each source, edition and page, every tolerance and the before-and-after values of every fix.

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A All comparisons

The tables below list every comparison in the order the tests record them. The kind column distinguishes printed values (*published*), formulas of the cited source evaluated in the test (*closed-form*) and exact solutions by another method (*independent*).

A.1 Analysis

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
cables	horizontal stay 50 m, H = 100 kN	Ernst (1965), Der E-Modul von Seilen; Irvine, Cable Structures (1981)	elongation (mm)	31.18	31.17	-0.013	closed-form
cables	horizontal stay 50 m, H = 100 kN	statics	tension (kN)	100	100	-0.000	closed-form
seismic	3-storey RC frame	ASCE 7-22 §12.8, Eqs. 12.8-1 to 12.8-7	T used (min(T1, Cu·Ta)) (s)	0.4789	0.4789	+0.000	closed-form
seismic	3-storey RC frame	ASCE 7-22 §12.8, Eqs. 12.8-1 to 12.8-7	Cs	0.125	0.125	+0.000	closed-form
seismic	3-storey RC frame	ASCE 7-22 §12.8, Eqs. 12.8-1 to 12.8-7	V_ELF = Cs·W (kN)	64.8	64.8	+0.000	closed-form
seismic	stiff 1-storey frame (T < T0)	ASCE 7-22 §12.8, Eqs. 12.8-1 to 12.8-7	T used (min(T1, Cu·Ta)) (s)	0.03808	0.03808	+0.000	closed-form
seismic	stiff 1-storey frame (T < T0)	ASCE 7-22 §12.8, Eqs. 12.8-1 to 12.8-7	Cs	0.125	0.125	+0.000	closed-form
seismic	stiff 1-storey frame (T < T0)	ASCE 7-22 §12.8, Eqs. 12.8-1 to 12.8-7	V_ELF = Cs·W (kN)	48.48	48.48	+0.000	closed-form
linear statics	propped cantilever, UDL	Hibbeler, Structural Analysis 10e, Ch. 10 (force method); Roark 9e Table 8.1 case 1c	R_B (kN)	22.5	22.5	-0.000	closed-form
linear statics	propped cantilever, UDL	Hibbeler, Structural Analysis 10e, Ch. 10 (force method); Roark 9e Table 8.1 case 1c	M_A (kN·m)	-45	-45	-0.000	closed-form
linear statics	fixed-fixed, centre load	Roark 9e Table 8.1 case 3a (fixed ends, centre load)	M_end (kN·m)	-37.5	-37.5	-0.000	closed-form
linear statics	fixed-fixed, centre load	Roark 9e Table 8.1 case 3a (fixed ends, centre load)	M_mid (kN·m)	37.5	37.5	+0.000	closed-form
linear statics	fixed-fixed, centre load	Roark 9e Table 8.1 case 3a (fixed ends, centre load)	δ_mid (mm)	0.4524	0.4524	+0.000	closed-form
linear statics	two-span continuous beam, UDL	Ghali & Neville, Structural Analysis 7e, Ch. 4 (three-moment equation)	M_B (kN·m)	-45	-45	-0.000	closed-form
linear statics	two-span continuous beam, UDL	Ghali & Neville, Structural Analysis 7e, Ch. 4 (three-moment equation)	R_B (kN)	75	75	+0.000	closed-form
linear statics	Gerber beam with hinge	STAAD.Pro Verification V7 type (Gerber beam), statics	R_C (kN)	30	30	+0.000	closed-form
linear statics	Gerber beam with hinge	STAAD.Pro Verification V7 type (Gerber beam), statics	M_A (kN·m)	-200	-200	-0.000	closed-form
linear statics	Gerber beam with hinge	STAAD.Pro Verification V7 type (Gerber beam), statics	M at hinge (kN·m)	0	0	+0.000	closed-form
modal	uniform cantilever, 32 elements	Clough & Penzien 3e, Table 18-1 (βnL of a cantilever)	f1 (lateral) (Hz)	8.26	8.26	+0.000	published
modal	uniform cantilever, 32 elements	Clough & Penzien 3e, Table 18-1 (βnL of a cantilever)	f2 (lateral) (Hz)	51.76	51.76	-0.000	published
modal	uniform cantilever, 32 elements	Clough & Penzien 3e, Table 18-1 (βnL of a cantilever)	f3 (lateral) (Hz)	144.9	144.9	-0.001	published
modal	two-storey shear building	Chopra, Dynamics of Structures 5e, Ch. 10 (two-storey shear frame)	f1 (Hz)	3.354	3.352	-0.061	closed-form

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
modal	two-storey shear building	Chopra, Dynamics of Structures 5e, Ch. 10 (two-storey shear frame)	f2 (Hz)	8.781	8.776	-0.054	closed-form
modal	two-storey shear building	Chopra, Dynamics of Structures 5e, Ch. 10 (two-storey shear frame)	first-mode effective mass ratio	0.9472	0.9468	-0.041	closed-form
seismic	two-storey shear building, flat spectrum	Chopra 5e Ch. 13 (RSA); Der Kiureghian (1981) CQC	base shear V_RSA (kN)	284.7	285.2	+0.155	closed-form
seismic	two-storey shear building, flat spectrum	seismic weight = D	W (kN)	600	600	+0.000	closed-form
moving loads	two 100 kN axles 2 m apart, 10 m span	Hibbeler, Structural Analysis 10e, Ch. 6 (influence lines, absolute maximum moment)	absolute max moment (kN·m)	405	405	-0.000	closed-form
moving loads	influence line of M at L/4	Hibbeler, Structural Analysis 10e, Ch. 6 (influence lines, absolute maximum moment)	peak ordinate (m)	1.875	1.875	-0.000	closed-form
moving loads	influence line of R_A	Hibbeler, Structural Analysis 10e, Ch. 6 (influence lines, absolute maximum moment)	ordinate at mid-span	0.5	0.5	-0.000	closed-form
nonlinear members	tension-only X brace	statics of a tension-only brace (compression diagonal buckled out)	tension diagonal (kN)	37.5	37.5	+0.000	closed-form
nonlinear members	tension-only X brace	statics of a tension-only brace (compression diagonal buckled out)	compression diagonal (kN)	0	0	+0.000	closed-form
nonlinear members	two-span beam, lift-off support	statics: the lifted support carries nothing, span 1 acts as a simple beam	R at the lifted support (kN)	0	0	+0.000	closed-form
nonlinear members	two-span beam, lift-off support	statics: the lifted support carries nothing, span 1 acts as a simple beam	R at the middle support (kN)	50	50	-0.000	closed-form
plastic collapse	simple beam, central load	Neal, The Plastic Methods of Structural Analysis 3e, Chs. 2–3	Pc (kN)	88.54	88.54	-0.000	closed-form
plastic collapse	propped cantilever, central load	Neal, The Plastic Methods of Structural Analysis 3e, Chs. 2–3	Pc (kN)	132.8	132.8	-0.000	closed-form
plastic collapse	fixed-ended beam, UDL	Neal, The Plastic Methods of Structural Analysis 3e, Chs. 2–3	wc (kN/m)	59.03	59.03	-0.000	closed-form
plastic collapse	two equal spans, UDL	Neal, The Plastic Methods of Structural Analysis 3e, Chs. 2–3	wc (kN/m)	43.01	43.01	-0.000	published
plastic collapse	fixed-base portal, V = 1.5H (combined mechanism)	Horne, Plastic Theory of Structures, Ch. 3; Neal 3e Ch. 3	λc	79.69	79.69	-0.000	closed-form
plates	SS square plate, n = 5	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	w_max (mm)	70.94	63.64	-10.286	published
plates	SS square plate, n = 5	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	Mx centre (kN·m/m)	7.664	7.47	-2.535	published
plates	clamped square plate, n = 5	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	w_max (mm)	22.01	18.26	-17.054	published
plates	clamped square plate, n = 5	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx centre (kN·m/m)	3.696	3.81	+3.091	published
plates	SS square plate, n = 9	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	w_max (mm)	70.94	68.74	-3.089	published
plates	SS square plate, n = 9	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	Mx centre (kN·m/m)	7.664	7.601	-0.826	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
plates	clamped square plate, n = 9	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	w_max (mm)	22.01	20.98	-4.678	published
plates	clamped square plate, n = 9	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx centre (kN·m/m)	3.696	3.709	+0.351	published
plates	clamped square plate, n = 9	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx mid-edge (extrapolated) (kN·m/m)	-8.208	-7.422	+9.575	published
plates	SS square plate, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	w_max (mm)	70.94	70.38	-0.790	published
plates	SS square plate, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	Mx centre (kN·m/m)	7.664	7.644	-0.256	published
plates	clamped square plate, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	w_max (mm)	22.01	21.83	-0.853	published
plates	clamped square plate, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx centre (kN·m/m)	3.696	3.677	-0.505	published
plates	clamped square plate, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx mid-edge (extrapolated) (kN·m/m)	-8.208	-7.953	+3.106	published
plates	SS square plate, n = 33	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	w_max (mm)	70.94	70.84	-0.131	published
plates	SS square plate, n = 33	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	Mx centre (kN·m/m)	7.664	7.657	-0.091	published
plates	clamped square plate, n = 33	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	w_max (mm)	22.01	22.07	+0.232	published
plates	clamped square plate, n = 33	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx centre (kN·m/m)	3.696	3.669	-0.741	published
plates	clamped square plate, n = 33	Timoshenko & Woinowsky-Krieger 2e, Table 35 (ν = 0.3)	Mx mid-edge (extrapolated) (kN·m/m)	-8.208	-8.136	+0.878	published
plates	SS plate b/a = 2, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	w_max (mm)	177	175.5	-0.852	published
plates	SS plate b/a = 2, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	Mx centre (short span) (kN·m/m)	16.27	16.23	-0.273	published
plates	SS plate b/a = 2, n = 17	Timoshenko & Woinowsky-Krieger 2e, Table 8 (ν = 0.3)	My centre (long span) (kN·m/m)	7.424	7.385	-0.525	published
plates	SS plate b/a = 2, n = 17	equilibrium	Σ reactions (kN)	320	320	+0.000	closed-form
linear statics	fixed-base portal, lateral load	Hibbeler 10e Ch. 11 (slope-deflection); Ghali & Neville 7e §3	sway (mm)	0.08889	0.0889	+0.012	closed-form
linear statics	3D L-frame, out-of-plane tip load	McGuire, Gallagher & Ziemian 2e, Ch. 13 (bending + torsion)	tip δy (mm)	28.38	28.38	-0.000	closed-form
linear statics	Pratt truss, 4 panels	Kassimali, Structural Analysis 6e, Ch. 4 (method of sections)	bottom chord, panel 2 (kN)	15	15	-0.000	closed-form
linear statics	Pratt truss, 4 panels	Kassimali, Structural Analysis 6e, Ch. 4 (method of sections)	top chord, panel 2 (kN)	-20	-20	+0.000	closed-form
linear statics	Pratt truss, 4 panels	Kassimali, Structural Analysis 6e, Ch. 4 (method of sections)	diagonal T4–B8 (kN)	7.071	7.071	-0.000	closed-form
section properties	rectangle b/t = 1	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1 = J/(b t^3), section FE	0.141	0.1405	-0.386	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
section properties	rectangle b/t = 1	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1, Pondus frame formula	0.141	0.1408	-0.118	published
section properties	rectangle b/t = 1.5	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1 = J/(b t^3), section FE	0.196	0.1956	-0.213	published
section properties	rectangle b/t = 1.5	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1, Pondus frame formula	0.196	0.1956	-0.185	published
section properties	rectangle b/t = 2	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1 = J/(b t^3), section FE	0.229	0.2284	-0.241	published
section properties	rectangle b/t = 2	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1, Pondus frame formula	0.229	0.2289	-0.052	published
section properties	rectangle b/t = 3	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1 = J/(b t^3), section FE	0.263	0.263	-0.012	published
section properties	rectangle b/t = 3	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1, Pondus frame formula	0.263	0.2634	+0.154	published
section properties	rectangle b/t = 5	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1 = J/(b t^3), section FE	0.291	0.2907	-0.098	published
section properties	rectangle b/t = 5	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1, Pondus frame formula	0.291	0.2913	+0.117	published
section properties	rectangle b/t = 10	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1 = J/(b t^3), section FE	0.312	0.3111	-0.277	published
section properties	rectangle b/t = 10	Timoshenko & Goodier, Theory of Elasticity 3e, §109 (torsion of rectangles)	k1, Pondus frame formula	0.312	0.3123	+0.107	published
section properties	I 300 x 150 x 6.5 x 9 (no fillets)	Timoshenko & Gere 2e §5.10: Cw = tf bf^3 $h0^2$/24	Cw (m^6 x 1e9)	107.2	107.1	-0.073	closed-form
section properties	thin channel 200 x 80 x 2	Timoshenko & Gere 2e §5.4; Roark 9e Table 9.? (thin-walled shear centre)	shear centre from web (mm)	28.24	28.22	-0.054	closed-form
section properties	H 300x150x6.5x9	JIS G 3192 (dimensions, mass and properties of H sections)	A (cm^2)	46.78	46.78	+0.002	published
section properties	H 300x150x6.5x9	JIS G 3192	Ix (cm^4)	7210	7209	-0.011	published
section properties	H 200x100x5.5x8	JIS G 3192 (dimensions, mass and properties of H sections)	A (cm^2)	26.67	26.67	-0.002	published
section properties	H 200x100x5.5x8	JIS G 3192	Ix (cm^4)	1810	1806	-0.241	published
section properties	H 400x200x8x13	JIS G 3192 (dimensions, mass and properties of H sections)	A (cm^2)	83.37	83.37	+0.001	published
section properties	H 400x200x8x13	JIS G 3192	Ix (cm^4)	2.35e+04	2.346e+04	-0.185	published
section properties	H 400x400x13x21	JIS G 3192 (dimensions, mass and properties of H sections)	A (cm^2)	218.7	218.7	-0.002	published
section properties	H 400x400x13x21	JIS G 3192	Ix (cm^4)	6.66e+04	6.662e+04	+0.031	published
section properties	H 700x300x13x24	JIS G 3192 (dimensions, mass and properties of H sections)	A (cm^2)	231.5	231.5	+0.018	published
section properties	H 700x300x13x24	JIS G 3192	Ix (cm^4)	1.97e+05	1.975e+05	+0.249	published
slab load transfer	3 x 8 m panel, one way	one-way slab (ly/lx > 2): load to the long edges, e.g. Hibbeler 10e §2.1	long beam load (kN)	120	120	-0.000	closed-form
slab load transfer	3 x 8 m panel, one way	one-way slab (ly/lx > 2): load to the long edges, e.g. Hibbeler 10e §2.1	short beam load (kN)	0	0	+0.000	closed-form

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
slab load transfer	4 x 6 m panel, 10 kPa	tributary areas by 45° lines (two-way slab on beams), e.g. Hibbeler 10e §2.1	long beam load (kN)	80	80	+0.000	closed-form
slab load transfer	4 x 6 m panel, 10 kPa	tributary areas by 45° lines (two-way slab on beams), e.g. Hibbeler 10e §2.1	short beam load (kN)	40	40	+0.000	closed-form
slab load transfer	4 x 6 m panel, 10 kPa	tributary areas by 45° lines (two-way slab on beams), e.g. Hibbeler 10e §2.1	long beam M_max (trapezoid) (kN·m)	76.67	76.67	+0.000	closed-form
slab load transfer	4 x 6 m panel, 10 kPa	tributary areas by 45° lines (two-way slab on beams), e.g. Hibbeler 10e §2.1	short beam M_max (triangle) (kN·m)	26.67	26.67	-0.000	closed-form
slab load transfer	4 x 6 m panel, 10 kPa	equilibrium	Σ reactions (kN)	240	240	+0.000	closed-form
stability	pinned-pinned column, 8 elements	Timoshenko & Gere, Theory of Elastic Stability 2e, Ch. 2	Pcr (kN)	7896	7896	+0.003	closed-form
stability	fixed-free column, 8 elements	Timoshenko & Gere, Theory of Elastic Stability 2e, Ch. 2	Pcr (kN)	1974	1974	+0.000	closed-form
stability	fixed-pinned column, 8 elements	Timoshenko & Gere, Theory of Elastic Stability 2e, Ch. 2	Pcr (kN)	1.615e+04	1.615e+04	+0.017	published
stability	fixed-fixed column, 8 elements	Timoshenko & Gere, Theory of Elastic Stability 2e, Ch. 2	Pcr (kN)	3.158e+04	3.16e+04	+0.051	closed-form
stability	fixed-base portal, sway buckling	Timoshenko & Gere 2e §2.5; AISC 360-16 Commentary Eq. C-A-7-2 (G_A = 0, G_B = 1)	Pcr per column (kN)	9224	9224	+0.001	closed-form
stability	cantilever P-Delta at 0.4 Pcr	Timoshenko & Gere 2e §1.11: $3(\tan u - u)/u^3$	sway amplification	1.658	1.658	-0.000	closed-form
staged construction	cantilever propped after its dead load	staged construction, superposition of the stages (closed form)	root moment (kN·m)	-202.5	-202.5	-0.001	closed-form
staged construction	cantilever propped after its dead load	staged construction, superposition of the stages (closed form)	prop force (kN)	-11.25	-11.25	+0.003	closed-form
supports and loads	cantilever on an elastic spring	STAAD.Pro Verification V8 type; compatibility	spring force (kN)	8.958	8.958	+0.000	closed-form
supports and loads	fixed-fixed beam, support settlement	Ghali & Neville 7e, Ch. 4; $6EI\Delta/L^2$	end moment	M	(kN·m)	207.2	207.2
supports and loads	fixed-fixed beam, temperature gradient	Ghali & Neville 7e, Ch. 6; $EI\alpha\Delta T/h$		M	at the ends (kN·m)	41.45	41.45
supports and loads	fixed-fixed beam, temperature gradient	Ghali & Neville 7e, Ch. 6; $EI\alpha\Delta T/h$		M	at mid-span (kN·m)	41.45	41.45
supports and loads	restrained bar, uniform temperature	Timoshenko, Strength of Materials; $EA\alpha\Delta T$	N (kN)	-1243	-1243	-0.000	closed-form
supports and loads	rigid diaphragm, two columns	stiffness distribution of storey shear	shear in column 1 (kN)	25	25	+0.000	closed-form
time history	undamped SDOF, step ground acceleration	Chopra, Dynamics of Structures 5e, §4.3 (DAF = 2)	peak relative displacement (mm)	42.67	42.67	+0.015	closed-form
time history	undamped SDOF, step ground acceleration	$\omega = \sqrt{(k/m)}$, $k = 3EI/h^3$	natural frequency (Hz)	1.079	1.079	-0.011	closed-form
linear statics	VM2 overhanging beam	ANSYS MAPDL Verification Manual VM2	bending stress (psi)	1.14e+04	1.14e+04	+0.035	published
linear statics	VM2 overhanging beam	ANSYS MAPDL Verification Manual VM2	mid-span deflection (up) (in)	0.182	0.1825	+0.255	published
linear statics	VM4 hinged bars	ANSYS MAPDL Verification Manual VM4	bar stress (psi)	1e+04	1e+04	+0.000	published
linear statics	VM4 hinged bars	ANSYS MAPDL Verification Manual VM4	apex deflection (in)	-0.12	-0.12	-0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
foundations	beam on Winkler foundation, βL = 10	Hetényi, Beams on Elastic Foundation (1946), infinite beam under a point load	w under the load (mm)	1.13	1.13	+0.015	closed-form
foundations	beam on Winkler foundation, βL = 10	Hetényi, Beams on Elastic Foundation (1946), infinite beam under a point load	M under the load (kN·m)	55.33	55.28	-0.089	closed-form

A.2 Steel and Eurocode design

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
AISC 360 tension	AISC D.1 W8x21	AISC Design Examples v16, D.1, p. D-3	ϕtPn, D2-1 yielding (kip)	277	277.2	+0.072	published
AISC 360 tension	AISC D.1 W8x21	AISC Design Examples v16, D.1, p. D-3	An (B4.3b, holes d + 1/8 in) (in^2)	4.76	4.76	+0.000	published
AISC 360 tension	AISC D.1 W8x21	AISC Design Examples v16, D.1, p. D-3	U (Table D3.1 case 2, $\bar{x}$ without fillets)	0.908	0.9071	-0.098	published
AISC 360 tension	AISC D.1 W8x21	AISC Design Examples v16, D.1, p. D-3	ϕtPn, D2-2 rupture (steel::check) (kip)	211	210.5	-0.240	published
AISC 360 compression	AISC E.1C W14x132	AISC Design Examples v16, E.1C, p. E-9	ϕcPn (E3-2) (kip)	892	893.4	+0.152	published
AISC 360 compression	AISC E.1C W14x132	AISC Design Examples v16, E.1C, p. E-9	ϕcFn (ksi)	23	23.02	+0.107	published
AISC 360 compression	AISC E.1D W14x90	AISC Design Examples v16, E.1D, p. E-11	ϕcPn (Lcx governs) (kip)	927	927.5	+0.056	published
AISC 360 compression	AISC E.1D W14x90	AISC Design Examples v16, E.1D, p. E-11	Fn (E3-2) (ksi)	38.9	38.89	-0.026	published
AISC 360 compression	INTI301 Ej.10(a) IPE300	Troglia et al. (2018), Ej. 10-2	Fcr = 0.877 Fe (E3-3) (MPa)	46	45.98	-0.038	published
AISC 360 compression	INTI301 Ej.10(a) IPE300	Troglia et al. (2018), Ej. 10-2	Pn (kN)	247.4	247.4	-0.006	published
AISC 360 flexure	AISC F.1-1B W18x50	AISC Design Examples v16, F.1-1B, p. F-8	Mp (F2-1) (kip-in)	5050	5050	-0.000	published
AISC 360 flexure	AISC F.1-1B W18x50	AISC Design Examples v16, F.1-1B, p. F-8	ϕbMn (kip-ft)	379	378.7	-0.066	published
AISC 360 flexure	AISC F.1-2B W18x50	AISC Design Examples v16, F.1-2B, p. F-10	Cb (F1-1), centre third	1.01	1.014	+0.348	published
AISC 360 flexure	AISC F.1-2B W18x50	AISC Design Examples v16, F.1-2B, p. F-11	Mn (F2-2) (kip-in)	4060	4067	+0.166	published
AISC 360 flexure	AISC F.1-2B W18x50	AISC Design Examples v16, F.1-2B, p. F-11	ϕbMn (kip-ft)	304	305	+0.331	published
AISC 360 flexure	AISC F.1-2B W18x50	AISC Design Examples v16, F.1-2B, p. F-10	Mn at the printed Lp = 69.9 in (= Mp) (kip-in)	5050	5050	-0.000	published
AISC 360 flexure	AISC F.1-2B W18x50	AISC Design Examples v16, F.1-2B, p. F-10	Mn at the printed Lr = 203 in (= 0.7 Fy Sx) (kip-in)	3112	3117	+0.177	published
AISC 360 flexure	AISC F.1-3B W18x50	AISC Design Examples v16, F.1-3B, p. F-14	Cb (F1-1), half span	1.3	1.299	-0.100	published
AISC 360 flexure	AISC F.1-3B W18x50	AISC Design Examples v16, F.1-3B, p. F-15	Fcr (F2-4) (ksi)	43.2	43.18	-0.043	published
AISC 360 flexure	AISC F.1-3B W18x50	AISC Design Examples v16, F.1-3B, p. F-15	Mn (F2-3) (kip-in)	3840	3839	-0.030	published
AISC 360 flexure	AISC F.1-3B W18x50	AISC Design Examples v16, F.1-3B, p. F-15	ϕbMn (kip-ft)	288	287.9	-0.030	published
AISC 360 flexure	AISC F.3B W21x48	AISC Design Examples v16, F.3B, p. F-25	Mn (F3-1, flange local buckling) (kip-in)	5310	5306	-0.072	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
AISC 360 flexure	AISC F.3B W21x48	AISC Design Examples v16, F.3B, p. F-25	ϕbMn (kip-ft)	398	398	-0.009	published
AISC 360 flexure	AISC F.5 W12x58	AISC Design Examples v16, F.5, p. F-29	Mny (F6-1) (kip-in)	1630	1625	-0.307	published
AISC 360 flexure	AISC F.5 W12x58	AISC Design Examples v16, F.5, p. F-29	ϕbMny (kip-ft)	122	121.9	-0.102	published
AISC 360 shear	AISC G.1B W24x62	AISC Design Examples v16, G.1B, p. G-4	ϕvVn = Vn (G2-1, ϕv = 1.0) (kip)	306	305.7	-0.088	published
AISC 360 flexure	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-9	Mn at the printed Lp = 8.76 ft (= Mp) (kip-in)	6950	6950	-0.000	published
AISC 360 flexure	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-9	Mn at the printed Lr = 33.2 ft (= 0.7 Fy Sx) (kip-in)	4305	4279	-0.601	published
AISC 360 flexure	AISC H.3 W14x82	F1-1 on the constructed diagram (target 1.41)	Cb	1.41	1.41	+0.000	closed-form
AISC 360 tension	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-9	ϕtPn (D2-1) (kip)	1080	1080	-0.000	published
AISC 360 flexure	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-10	Mnx (F2-2, Cb 1.41) (kip-in)	6560	6538	-0.331	published
AISC 360 flexure	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-10	ϕbMnx (kip-ft)	492	490.4	-0.331	published
AISC 360 flexure	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-10	Mny (F6-1) (kip-in)	2240	2240	-0.000	published
AISC 360 flexure	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-10	ϕbMny (kip-ft)	168	168	-0.000	published
AISC 360 interaction	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-11	Pr/Pc (steel::check)	0.161	0.1611	+0.069	published
AISC 360 interaction	AISC H.3 W14x82	AISC Design Examples v16, H.3, p. H-11	H1-1b (steel::check)	0.873	0.8745	+0.169	published
EN 1992-1-1 (cross-check)	JRC-EC2 3.2.1.3 T-beam	Biasioli et al. (2014), §3.2.1.3, p. 57	MRd(As = 556 mm^2), mid-span (kNm)	89.3	89.26	-0.050	published
EN 1992-1-1 (cross-check)	JRC-EC2 3.2.1.3 T-beam	Biasioli et al. (2014), §3.2.1.3, p. 58	MRd(As = 947 mm^2), support (kNm)	132.9	132.8	-0.058	published
EN 1992-1-1 (cross-check)	JRC-EC2 3.2.1.4 shear	Biasioli et al. (2014), §3.2.1.4, p. 59	VRd,c (6.2.a) (kN)	47.9	47.9	+0.005	published
EN 1992-1-1 (cross-check)	JRC-EC2 3.2.1.4 shear	Biasioli et al. (2014), §3.2.1.4, p. 60	VRd,s (6.8, asw 317 mm^2/m, cot θ 2.5) (kN)	115.5	115.4	-0.138	published
EN 1992-1-1 (cross-check)	stirrup grade	EN 1992-1-1 (6.8): VRd,s $\propto$ fywd	VRd,s(fyt 235) / VRd,s(fyt 390)	0.6026	0.6026	+0.000	closed-form
EN 1993-1-1	H 300x150x6.5x9, M20, 2 holes per flange	EN 1993-1-1 6.2.3(2)b: 0.9 Anet fu/γM2	Nu,Rd (kN)	1119	1119	+0.000	closed-form
EN 1993-1-1	P364 Ex.9 UKC 356x368x129	SCI P364 (2009), Ex. 9, sheet 2	section class (Table 5.2)	3	3	+0.000	published
EN 1993-1-1	P364 Ex.9 UKC 356x368x129	SCI P364 (2009), Ex. 9, sheet 4 (χ to 2 dp)	χz	0.65	0.649	-0.148	published
EN 1993-1-1	P364 Ex.9 UKC 356x368x129	SCI P364 (2009), Ex. 9, sheet 4	Nb,z,Rd (kN)	3678	3672	-0.156	published
EN 1993-1-1	P364 Ex.9 UKC 356x368x129	SCI P364 (2009), Ex. 9 (Blue Book)	Nb,y,Rd (kN)	5010	5012	+0.047	published
EN 1993-1-1	P364 Ex.9 UKC 356x368x129	SCI P364 (2009), Ex. 9, sheet 3	Nc,Rd = A fy/γM0 (kN)	5658	5658	+0.000	published
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-59	Av (6.2.6(3)) (mm^2)	3080	3080	+0.008	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-58	Mc,Rd (6.2.5) (kNm)	189	189	+0.000	published
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-58	MEd/Mc,Rd	0.479	0.4787	-0.062	published
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-59	Vpl,Rd (6.2.6) (kN)	417.9	417.9	+0.005	published
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-60 (SN003)	C1 (Pondus: SN003 value for the diagram it recognises)	1.127	1.127	+0.000	published
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-61	Mcr (top flange, zg = +165 mm) (kNm)	113.9	113.9	-0.037	published
EN 1993-1-1	SBE A.1 IPE330	SBE Multi-Storey Part 4 (2010), A.1, p. 4-61	$\bar{\lambda}$LT	1.288	1.288	+0.033	published
EN 1993-1-1	SBE A.2 IPE400	SBE Multi-Storey Part 4 (2010), A.2, p. 4-66	Av (6.2.6(3)) (mm^2)	4269	4269	+0.002	published
EN 1993-1-1	SBE A.2 IPE400	SBE Multi-Storey Part 4 (2010), A.2, p. 4-66	Mc,Rd (6.2.5) (kNm)	307.1	307.1	-0.002	published
EN 1993-1-1	SBE A.2 IPE400	SBE Multi-Storey Part 4 (2010), A.2, p. 4-66	Vpl,Rd (6.2.6) (kN)	579.2	579.2	+0.002	published
EN 1993-1-1	SBE A.5 HEB300	SBE Multi-Storey Part 4 (2010), A.5, p. 4-96	χy	0.808	0.8083	+0.034	published
EN 1993-1-1	SBE A.5 HEB300	SBE Multi-Storey Part 4 (2010), A.5, p. 4-96	χz	0.671	0.6705	-0.075	published
EN 1993-1-1	SBE A.5 HEB300	SBE Multi-Storey Part 4 (2010), A.5, p. 4-97	Nb,Rd (6.3.1.1) (kN)	2350	2348	-0.074	published
EN 1993-1-1	SBE A.5 HEB300	SBE Multi-Storey Part 4 (2010), A.5, p. 4-97	NEd/Nb,Rd (printed to 2 figures)	0.85	0.8519	+0.221	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	uniform moment: C1 = 1 (SN003)	C1	1	1	+0.000	closed-form
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, p. 8-18	section class under N + M (Table 5.2)	1	1	+0.000	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, pp. 8-19 – 8-21	χy	1	1	+0.000	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, pp. 8-19 – 8-21	χz	0.865	0.866	+0.112	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, pp. 8-19 – 8-21	Nb,y,Rd (kN)	3507	3507	+0.011	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, pp. 8-19 – 8-21	Nb,z,Rd (kN)	3034	3037	+0.109	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, pp. 8-19 – 8-21	kyy (Annex B, Table B.2)	0.997	0.9971	+0.011	published
EN 1993-1-1	SBE-MS8 Ex.2 IPE450	SBE Multi-Storey Part 8 (2010), App. A Ex. 2, pp. 8-19 – 8-21	kzy (Annex B, Table B.2)	0.997	0.997	-0.001	published
Connections	shear tab 4-M20 (Fnv 372 MPa) at 75 mm	AISC 360-16 J3.6, elastic bolt-group method	bolt ratio	0.6683	0.6683	-0.000	closed-form
Connections	shear tab 4-M20 (Fnv 372 MPa) at 75 mm	AISC 360-16 J3.10 (2.4 d t Fu, web 8 mm)	bearing ratio	0.5085	0.5085	-0.000	closed-form
Connections	shear tab 4-M20 (Fnv 372 MPa) at 75 mm	AISC 360-16 J4.2 (yielding, rupture)	plate shear ratio	0.384	0.384	+0.000	closed-form

A.3 Concrete, timber and masonry design

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
RC column biaxial	Pincheira 10.20.1	StructurePoint Biaxial Rectangular Column (ACI 318-19) after Pincheira Ex. 10.20.1	Mnx by load contour α 1.15 vs exact (kip-ft)	321	294.6	-8.225	independent
RC column biaxial	Pincheira 10.20.1	StructurePoint Biaxial Rectangular Column (ACI 318-19) after Pincheira Ex. 10.20.1	Mny by load contour α 1.15 vs exact (kip-ft)	201	184.1	-8.396	independent

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	Aoh (mm^2)	1.389e+05	1.389e+05	-0.000	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	ph (mm)	1575	1575	-0.000	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	Ao = 0.85 Aoh (mm^2)	1.18e+05	1.18e+05	+0.001	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	ϕTth (SI 0.083$\sqrt{f'c}$) (kN·m)	7.959	7.958	-0.010	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	ϕTcr = 4 ϕTth (kN·m)	31.85	31.83	-0.053	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	At/s per leg (mm^2/mm)	0.5182	0.5183	+0.024	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	Al (22.7.6.1b) (mm^2)	816.1	816.2	+0.008	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	Al,min (9.6.4.3; SI 0.42$\sqrt{f'c}$ for 5$\sqrt{f'c}$: +1.1 % on a difference) (mm^2)	326.5	339.6	+4.023	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	ϕVc (SI 0.17: +2.4 %) (kN)	127	130	+2.350	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	(Av + 2At)/s min (SI 0.35 for 50 psi) (mm^2/mm)	0.2972	0.3009	+1.237	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	s,max = ph/8 (9.7.6.3.3) (mm)	196.8	196.8	-0.000	published
RC torsion	Wight 7-2	StructurePoint Equilibrium Torsion (ACI 318-14) after Wight Ex. 7-2	cross-section limit 325.55 / 474.34 psi	0.6863	0.6866	+0.044	published
RC beam flexure	2.I.6 singly reinforced	INTI-CIRSOC 201 Ej. 2.I.6 (2005), as recorded in Stabileo	ϕMn (ϕ = 0.90, Mn = 56.20) (kN·m)	50.58	50.59	+0.010	published
RC beam flexure	2.I.6 singly reinforced	INTI-CIRSOC 201 Ej. 2.I.6 (2005), as recorded in Stabileo	kc = c/d with β1(25 MPa)	0.213	0.2128	-0.079	published
masonry TMS 402	out-of-plane flexure, d = t/2	TMS 402-16 9.3.2 (Whitney block 0.80 f'm)	Mu / ϕMn	0.6926	0.6926	-0.000	closed-form
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.033	0.033	+0.000	published
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.04	0.04	+0.000	published
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.048	0.048	+0.000	published
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.055	0.055	+0.000	published
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.063	0.063	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.083	0.083	+0.000	published
two-way slab coefficients	case 1 interior, neg. continuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.033	0.033	+0.000	published
two-way slab coefficients	case 1 interior, neg. continuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.083	0.083	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.025	0.025	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.03	0.03	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.036	0.036	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.041	0.041	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.047	0.047	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.062	0.062	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.025	0.025	+0.000	published
two-way slab coefficients	case 1 interior, pos. midspan, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.062	0.062	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.041	0.041	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.048	0.048	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.055	0.055	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.062	0.062	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.069	0.069	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.085	0.085	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.041	0.041	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. continuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.085	0.085	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.021	0.021	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.024	0.024	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.027	0.027	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.031	0.031	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.035	0.035	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.042	0.042	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.021	0.021	+0.000	published
two-way slab coefficients	case 2 one edge disc., neg. discontinuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.042	0.042	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.031	0.031	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.036	0.036	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.041	0.041	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.047	0.047	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.052	0.052	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.064	0.064	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.031	0.031	+0.000	published
two-way slab coefficients	case 2 one edge disc., pos. midspan, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.064	0.064	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.049	0.049	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.057	0.057	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.064	0.064	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.071	0.071	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.078	0.078	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.09	0.09	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.049	0.049	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. continuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.09	0.09	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.025	0.025	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.028	0.028	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.032	0.032	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.036	0.036	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.039	0.039	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.045	0.045	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.025	0.025	+0.000	published
two-way slab coefficients	case 3 two edges disc., neg. discontinuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.045	0.045	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.037	0.037	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.043	0.043	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.048	0.048	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.054	0.054	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.059	0.059	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.068	0.068	+0.000	published
two-way slab coefficients	case 3 two edges disc., pos. midspan, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.037	0.037	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 3 two edges disc., pos. midspan, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.068	0.068	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.058	0.058	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.066	0.066	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.074	0.074	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.082	0.082	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.09	0.09	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.098	0.098	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.058	0.058	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. continuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.098	0.098	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.029	0.029	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.033	0.033	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.037	0.037	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.041	0.041	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.045	0.045	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.049	0.049	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.029	0.029	+0.000	published
two-way slab coefficients	case 4 three edges disc., neg. discontinuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.049	0.049	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.044	0.044	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.05	0.05	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.056	0.056	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.062	0.062	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.068	0.068	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.074	0.074	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.044	0.044	+0.000	published
two-way slab coefficients	case 4 three edges disc., pos. midspan, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.074	0.074	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.033	0.033	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.038	0.038	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.043	0.043	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.047	0.047	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.053	0.053	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.055	0.055	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.033	0.033	+0.000	published
two-way slab coefficients	case 5 four edges disc., neg. discontinuous edge, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.055	0.055	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 1	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.05	0.05	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 0.9	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.057	0.057	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 0.8	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.064	0.064	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 0.7	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.072	0.072	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 0.6	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.08	0.08	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 0.5	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.083	0.083	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, long span	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.05	0.05	+0.000	published
two-way slab coefficients	case 5 four edges disc., pos. midspan, short span, m = 0.4 (0.5 and less)	ACI 318-63 App. A Method 2 Table I (scan in Hussain, U. Basrah notes)	C	0.083	0.083	+0.000	published
two-way slab coefficients	Ex. 1 interior, m = 0.833, short neg.	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	C	0.04536	0.04536	+0.000	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
two-way slab coefficients	Ex. 1 interior, m = 0.833, short pos.	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	C	0.03402	0.03402	+0.000	published
two-way slab coefficients	Ex. 1 interior, long neg.	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	C	0.033	0.033	+0.000	published
two-way slab design	Ex. 1 interior	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	wu (kPa)	25.36	25.36	+0.000	published
two-way slab design	Ex. 1 interior	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	short-span negative Mu (kN·m/m)	41.41	41.39	-0.054	published
two-way slab design	Ex. 1 interior	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	short-span positive Mu (kN·m/m)	31.06	31.04	-0.062	published
two-way slab design	Ex. 1 interior	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	long-span negative Mu (kN·m/m)	30.13	30.13	-0.008	published
two-way slab design	Ex. 1 interior	Hussain, U. Basrah two-way slab notes, Method 2 Example 1 (interior panel 6.0 x 7.2 m)	long-span positive Mu (kN·m/m)	22.82	22.82	+0.018	published
two-way slab design	Ex. 2 corner	Hussain, U. Basrah two-way slab notes, Method 2 Example 2 (corner panel 6.1 x 6.1 m)	wu (kPa)	11.72	11.72	-0.000	published
two-way slab design	Ex. 2 corner	Hussain, U. Basrah two-way slab notes, Method 2 Example 2 (corner panel 6.1 x 6.1 m)	negative Mu, continuous edge (C 0.049) (kN·m/m)	21.37	21.37	-0.005	published
two-way slab design	Ex. 2 corner	Hussain, U. Basrah two-way slab notes, Method 2 Example 2 (corner panel 6.1 x 6.1 m)	negative Mu, discontinuous edge (C 0.025) (kN·m/m)	10.9	10.9	+0.023	published
two-way slab design	Ex. 2 corner	Hussain, U. Basrah two-way slab notes, Method 2 Example 2 (corner panel 6.1 x 6.1 m)	positive Mu (C 0.037) (kN·m/m)	16.14	16.14	-0.026	published
one-way slab	end span	StructurePoint One-Way Slab (ACI 318-14) after Wight Ex. 5-7, Tables 1-2	Mu at the exterior support, $wl^2/24$ (kN·m/m)	8.14	8.149	+0.112	published
one-way slab	end span	StructurePoint One-Way Slab (ACI 318-14) after Wight Ex. 5-7, Tables 1-2	Vu at the first interior support, 1.15 wl/2 (kN/m)	28.17	28.21	+0.150	published
one-way slab	interior span	StructurePoint One-Way Slab (ACI 318-14) after Wight Ex. 5-7, Tables 1-2	Mu at midspan, $wl^2/16$ (kN·m/m)	13.66	13.67	+0.073	published
one-way slab	interior span	StructurePoint One-Way Slab (ACI 318-14) after Wight Ex. 5-7, Tables 1-2	Mu at the supports, $wl^2/11$ (kN·m/m)	19.88	19.88	-0.028	published
one-way slab	interior span	StructurePoint One-Way Slab (ACI 318-14) after Wight Ex. 5-7, Tables 1-2	Vu at the supports, wl/2 (kN/m)	25.98	25.94	-0.160	published
RC beam flexure	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	As required (mm^2)	1853	1853	+0.018	published
RC beam flexure	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	$\beta 1$ at 4.35 ksi (SI form of Table 22.2.2.4.3; printed to 2 digits)	0.83	0.8358	+0.695	published
RC beam flexure	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	As,min (9.6.1.2b governs; SI 1.4/fy for 200/fy) (mm^2)	452.9	460.1	+1.584	published
RC beam shear	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	ϕVc (SI 0.17 for $2\sqrt{f'c}$ psi: +2.4 %) (kN)	92.75	94.93	+2.352	published
RC beam shear	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	Vs required (inherits the ϕVc offset: −6.4 %) (kN)	45.51	42.58	-6.423	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
RC beam shear	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	(Av/s)min (SI 0.35 bw/fyt for 50 bw/fyt) (mm^2/mm)	0.254	0.2579	+1.526	published
RC beam shear	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	s,max = d/2 (mm)	223	223	+0.000	published
RC beam deflection	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	kd (mm)	161.8	161.8	-0.013	published
RC beam deflection	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	Icr (mm^4)	1.565e+09	1.564e+09	-0.044	published
RC beam deflection	Chaallal SS beam	StructurePoint Simply Supported RC Beam (ACI 318-14) after Chaallal (2018)	Mcr (SI 0.62$\sqrt{}$f'c for 7.5$\sqrt{}$f'c) (kN·m)	44.71	44.51	-0.452	published
RC beam deflection	Chaallal SS beam, Ie under D	ACI 318-19 Table 24.2.3.5 on the example's Icr, Ig, Mcr, Ma	Ie (in^4)	4009	4009	-0.000	closed-form
RC beam deflection	Chaallal SS beam, Ie under D + L	ACI 318-19 Table 24.2.3.5 on the example's Icr, Ig, Mcr, Ma	Ie (in^4)	3807	3807	-0.000	closed-form
RC column slender-ness	Wang 13.17.3	ACI 318-19 6.6.4.4.2 with (EI)eff 6.6.4.4.4(a), βdns 0.4	Pc (option a) (kip)	4627	4627	+0.000	closed-form
RC column slender-ness	Wang 13.17.3	StructurePoint Slender Column Non-Sway (ACI 318-19) after Wang Ex. 13.17.3	Mc = δns M2 (kip-ft)	105	105	+0.000	published
RC column slender-ness	Wang 13.17.3	StructurePoint Slender Column Non-Sway (ACI 318-19) after Wang Ex. 13.17.3	δns	1	1	+0.000	published
RC column slender-ness	Wang 13.17.3	StructurePoint Slender Column Non-Sway (ACI 318-19) after Wang Ex. 13.17.3	M2,min = Pu (15 + 0.03h) (SI 15 mm for 0.6 in) (kip-ft)	48.56	48.15	-0.846	published
RC column slender-ness	Wang 13.17.3 with 1.8 klu	ACI 318-19 6.6.4.5.2-3, Cm = 0.6 + 0.4 M1/M2	δns	1.177	1.177	+0.000	closed-form
RC footing	Wight 15-2	StructurePoint Square Spread Footing (ACI 318-14) after Wight Ex. 15-2	Mu at the column face over the width (kip-ft)	954	953.7	-0.026	published
RC footing	Wight 15-2	StructurePoint Square Spread Footing (ACI 318-14) after Wight Ex. 15-2	As over the width (in^2)	7.76	7.758	-0.024	published
RC footing	Wight 15-2	StructurePoint Square Spread Footing (ACI 318-14) after Wight Ex. 15-2	As,min = 0.0018 b h over the width (in^2)	7.72	7.721	+0.009	published
RC column P-M	tied 16x16	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	P0 (kip)	1534	1534	-0.000	published
RC column P-M	tied 16x16	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕPn,max = 0.80 ϕ P0 (kip)	797.7	797.7	-0.003	published
RC column P-M	tied 16x16, fs = 0	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Pn (kip)	957.4	960.9	+0.360	published
RC column P-M	tied 16x16, fs = 0	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Mn (kip-ft)	261.3	260.5	-0.311	published
RC column P-M	tied 16x16, fs = 0.5 fy	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Pn (kip)	649.1	651.8	+0.413	published
RC column P-M	tied 16x16, fs = 0.5 fy	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Mn (kip-ft)	338.5	338.5	-0.009	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
RC column P-M	tied 16x16, fs = fy (balanced)	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Pn (kip)	416.8	418.9	+0.498	published
RC column P-M	tied 16x16, fs = fy (balanced)	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Mn (kip-ft)	385.8	386.1	+0.077	published
RC column P-M	tied 16x16, εt = εy + 0.003	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Pn (kip)	190.7	192.1	+0.739	published
RC column P-M	tied 16x16, εt = εy + 0.003	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Mn (kip-ft)	318.6	319.1	+0.149	published
RC column P-M	tied 16x16, pure bending	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	Mn at Pn = 0 (kip-ft)	237.7	237.7	-0.023	published
RC column P-M	tied 16x16, pure bending	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕ (εt 0.00946 > εy + 0.003)	0.9	0.9	+0.000	published
RC column P-M	tied 16x16, fs = 0	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕMn at the printed ϕPn (kip-ft)	169.9	170	+0.099	published
RC column P-M	tied 16x16, fs = 0.5 fy	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕMn at the printed ϕPn (kip-ft)	220.1	220.4	+0.162	published
RC column P-M	tied 16x16, fs = fy (balanced)	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕMn at the printed ϕPn (kip-ft)	250.8	251.7	+0.373	published
RC column P-M	tied 16x16, εt = εy + 0.003	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕMn at the printed ϕPn (kip-ft)	286.8	286.7	-0.024	published
RC column P-M	tied 16x16, pure bending	StructurePoint Interaction Diagram Tied Column (ACI 318-19), points 1-6	ϕMn at the printed ϕPn (kip-ft)	214	213.9	-0.024	published
timber NDS	column CP	Dlubal KB 001674, NDS 2018 timber column	CP (NDS 3.7-1, c = 0.8; FcE 1 342 psi)	0.866	0.8666	+0.074	published
timber NDS	column CP	Dlubal KB 001674, NDS 2018 timber column	η = fc / F'c	0.913	0.9139	+0.095	published
Wood-Armer	normal, both faces	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	13	13	+0.000	closed-form
Wood-Armer	normal, both faces	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	8	8	+0.000	closed-form
Wood-Armer	normal, both faces	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	normal, both faces	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	bottom x special	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	bottom x special	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	6.4	6.4	+0.000	closed-form
Wood-Armer	bottom x special	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	10.67	10.67	+0.000	closed-form
Wood-Armer	bottom x special	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	bottom y special	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	6.4	6.4	+0.000	closed-form
Wood-Armer	bottom y special	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	bottom y special	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	bottom y special	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	10.67	10.67	+0.000	closed-form
Wood-Armer	top x special	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	13	13	+0.000	closed-form
Wood-Armer	top x special	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	1	1	+0.000	closed-form

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
Wood-Armer	top x special	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	top x special	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	2.9	2.9	+0.000	closed-form
Wood-Armer	top y special	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	1	1	+0.000	closed-form
Wood-Armer	top y special	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	13	13	+0.000	closed-form
Wood-Armer	top y special	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	2.9	2.9	+0.000	closed-form
Wood-Armer	top y special	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	pure twist	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	4	4	+0.000	closed-form
Wood-Armer	pure twist	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	4	4	+0.000	closed-form
Wood-Armer	pure twist	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	4	4	+0.000	closed-form
Wood-Armer	pure twist	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	4	4	+0.000	closed-form
Wood-Armer	hogging both	Wood (1968) / Armer (1968), normal and special rules	Mx* bottom (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	hogging both	Wood (1968) / Armer (1968), normal and special rules	My* bottom (kN·m/m)	0	0	+0.000	closed-form
Wood-Armer	hogging both	Wood (1968) / Armer (1968), normal and special rules	Mx* top (kN·m/m)	14.5	14.5	+0.000	closed-form
Wood-Armer	hogging both	Wood (1968) / Armer (1968), normal and special rules	My* top (kN·m/m)	10.5	10.5	+0.000	closed-form

A.4 Shear walls under earthquake and dynamic loading

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
wall modal	cantilever wall 0.3 × 2 m, H 60 m, 20 storeys	Clough & Penzien 3e, Table 18-1 (βnL of a uniform cantilever)	f1 (in plane) (Hz)	0.2753	0.2751	-0.077	published
wall modal	cantilever wall 0.3 × 2 m, H 60 m, 20 storeys	Clough & Penzien 3e, Table 18-1 (βnL of a uniform cantilever)	f2 (in plane) (Hz)	1.725	1.718	-0.444	published
wall modal	cantilever wall 0.3 × 2 m, H 60 m, 20 storeys	Clough & Penzien 3e, Table 18-1 (βnL of a uniform cantilever)	f3 (in plane) (Hz)	4.831	4.781	-1.028	published
wall modal	cantilever wall 0.3 × 2 m, H 60 m, 20 storeys	Chopra 5e Ch. 17 (uniform cantilever, $(\int m\phi)^2/\int m\phi^2$)	effective mass ratio, mode 1	0.6131	0.6133	+0.043	closed-form
wall modal	cantilever wall 0.3 × 2 m, H 60 m, 20 storeys	Chopra 5e Ch. 17 (uniform cantilever, $(\int m\phi)^2/\int m\phi^2$)	effective mass ratio, mode 2	0.1883	0.1889	+0.318	closed-form
wall modal	cantilever wall 0.3 × 2 m, H 60 m, 20 storeys	Chopra 5e Ch. 17 (uniform cantilever, $(\int m\phi)^2/\int m\phi^2$)	effective mass ratio, mode 3	0.06473	0.0652	+0.720	closed-form
torsional coupling	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	Chopra 5e Ch. 9 (one-storey unsymmetric-plan system), 3 × 3 eigenproblem	f1 (Hz)	3.524	3.524	-0.001	closed-form
torsional coupling	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	Chopra 5e Ch. 9 (one-storey unsymmetric-plan system), 3 × 3 eigenproblem	f2 (Hz)	4.279	4.279	-0.012	closed-form
torsional coupling	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	Chopra 5e Ch. 9 (one-storey unsymmetric-plan system), 3 × 3 eigenproblem	f3 (Hz)	47.66	47.65	-0.031	closed-form

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
torsional coupling	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	Chopra 5e Ch. 9 (one-storey unsymmetric-plan system), 3 × 3 eigenproblem	mass ratio in y, mode 1	0.5946	0.5944	-0.043	closed-form
torsional coupling	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	Chopra 5e Ch. 9 (one-storey unsymmetric-plan system), 3 × 3 eigenproblem	mass ratio in y, mode 3	0.4054	0.4054	+0.006	closed-form
wall RSA	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	Chopra 5e Ch. 13; Der Kiureghian (1981) CQC, ξ = 5 %	base shear V_RSA in y (flat Sa 0.4 g) (kN)	86.38	86.39	+0.018	closed-form
wall RSA	one storey, wall on one side (x = 0), mass centre at x = 6.67 m	the loaded edges' weight	seismic weight W (kN)	300	300	+0.000	closed-form
wall-frame modal	wall + shear frames, αH = 0.5, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (in plane) (Hz)	0.2879	0.2874	-0.178	independent
wall-frame modal	wall + shear frames, αH = 0.5, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f2 (in plane) (Hz)	1.74	1.73	-0.540	independent
wall-frame modal	wall + shear frames, αH = 0.5, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f3 (in plane) (Hz)	4.843	4.789	-1.124	independent
wall-frame modal	wall + shear frames, αH = 0.5, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	effective mass ratio, mode 1	0.6156	0.6159	+0.048	independent
wall-frame modal	wall + shear frames, αH = 1, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (in plane) (Hz)	0.3218	0.3212	-0.187	independent
wall-frame modal	wall + shear frames, αH = 1, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f2 (in plane) (Hz)	1.782	1.772	-0.533	independent
wall-frame modal	wall + shear frames, αH = 1, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f3 (in plane) (Hz)	4.88	4.825	-1.117	independent
wall-frame modal	wall + shear frames, αH = 1, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	effective mass ratio, mode 1	0.6227	0.623	+0.056	independent
wall-frame modal	wall + shear frames, αH = 3, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (in plane) (Hz)	0.5419	0.5406	-0.237	independent
wall-frame modal	wall + shear frames, αH = 3, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f2 (in plane) (Hz)	2.173	2.162	-0.496	independent

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
wall-frame modal	wall + shear frames, αH = 3, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f3 (in plane) (Hz)	5.253	5.198	-1.054	independent
wall-frame modal	wall + shear frames, αH = 3, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	effective mass ratio, mode 1	0.6692	0.6699	+0.101	independent
wall-frame modal	wall + shear frames, αH = 8, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (in plane) (Hz)	1.138	1.134	-0.332	independent
wall-frame modal	wall + shear frames, αH = 8, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f2 (in plane) (Hz)	3.743	3.724	-0.506	independent
wall-frame modal	wall + shear frames, αH = 8, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f3 (in plane) (Hz)	7.256	7.19	-0.902	independent
wall-frame modal	wall + shear frames, αH = 8, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	effective mass ratio, mode 1	0.7351	0.7363	+0.177	independent
wall-frame modal	wall + shear frames, αH = 8, 10 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (convergence with the storeys) (Hz)	1.138	1.131	-0.611	independent
wall-frame modal	wall + shear frames, αH = 8, 20 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (convergence with the storeys) (Hz)	1.138	1.134	-0.350	independent
wall-frame modal	wall + shear frames, αH = 8, 40 storeys	Heidebrecht & Stafford Smith (1973); Miranda & Taghavi (2005): wall-frame continuum, frequency equation solved independently	f1 (convergence with the storeys) (Hz)	1.138	1.134	-0.332	independent
wall time history	cantilever wall 0.3 × 2 m, H 60 m, step a0 = 0.1 g, undamped	Chopra 5e Ch. 4 and 13: step response by modal superposition (6 modes)	peak top displacement (mm)	1026	1028	+0.187	closed-form
wall time history	cantilever wall 0.3 × 2 m, H 60 m, step a0 = 0.1 g, undamped	the same, as a dynamic amplification of the static top displacement	peak / (m a0 H^4/8EI)	2.025	2.029	+0.187	closed-form

A.5 Bridge under moving load

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.1L (kip-ft)	485	485.2	+0.042	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.1L (kip-ft)	846	846	-0.005	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.1L (kip-ft)	-60	-59.93	+0.123	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.1L (kip-ft)	-112	-112.3	-0.311	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.2L (kip-ft)	816	815.6	-0.054	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.2L (kip-ft)	1434	1434	+0.008	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.2L (kip-ft)	-120	-119.9	+0.123	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.2L (kip-ft)	-225	-224.7	+0.134	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.3L (kip-ft)	1002	1002	-0.002	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.3L (kip-ft)	1779	1779	+0.000	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.3L (kip-ft)	-180	-179.8	+0.123	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.3L (kip-ft)	-337	-337	-0.014	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.4L (kip-ft)	1083	1083	-0.020	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.4L (kip-ft)	1932	1932	-0.021	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.4L (kip-ft)	-240	-239.7	+0.123	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.4L (kip-ft)	-449	-449.4	-0.088	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.5L (kip-ft)	1059	1059	+0.046	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.5L (kip-ft)	1894	1894	-0.006	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.5L (kip-ft)	-300	-299.6	+0.123	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.5L (kip-ft)	-562	-561.7	+0.045	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.6L (kip-ft)	951	950.9	-0.013	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.6L (kip-ft)	1691	1691	-0.008	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.6L (kip-ft)	-359	-359.6	-0.155	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.6L (kip-ft)	-674	-674.1	-0.014	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.7L (kip-ft)	743	743.3	+0.045	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.7L (kip-ft)	1304	1304	+0.033	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.7L (kip-ft)	-419	-419.5	-0.116	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.7L (kip-ft)	-786	-786.4	-0.056	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.8L (kip-ft)	463	463.3	+0.067	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.8L (kip-ft)	770	769.8	-0.029	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.8L (kip-ft)	-479	-479.4	-0.086	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.8L (kip-ft)	-899	-898.8	+0.020	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M+ at 0.9L (kip-ft)	192	192	-0.020	published

Aspect	Case	Source	Quantity (unit)	Reference	Pondus	Error (%)	Kind
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	1.33 vehicle + lane M+ at 0.9L (kip-ft)	285	286.8	+0.646	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 0.9L (kip-ft)	-539	-539.3	-0.062	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 0.9L (kip-ft)	-1102	-1103	-0.110	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	special M- (90 % of two trucks + lane) at 0.9L (kip-ft)	-1250	-1252	-0.136	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	vehicle M- at 1.0L (kip-ft)	-599	-599.3	-0.044	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	standard M- (1.33 vehicle + lane) at 1.0L (kip-ft)	-1450	-1450	+0.003	published
bridge moving load	two-span 90 ft + 90 ft composite I-girder, HL-93 per lane	FHWA Steel Bridge Design Handbook, Design Example 2A (2015), Table 6	special M- (90 % of two trucks + lane) at 1.0L (kip-ft)	-2020	-2020	-0.007	published